

SISTEMAS ELECTRÓNICOS DE ALIMENTACIÓN

BLOQUE TEMÁTICO II ALIMENTACIÓN DE SISTEMAS ELECTRÓNICOS

TEMA 4

CONVERTIDORES CONMUTADOS DC/DC

Convertidores Conmutados DC/DC

- Comparación entre alimentación lineal y conmutada
- Topologías de convertidores no aislados. Análisis, formas de onda, parámetros. Funcionamiento en modos continuo, discontinuo
- Control PWM en modo tensión y en modo corriente. Circuitos integrados comerciales
- Soluciones integradas de alimentación conmutada. Diseño

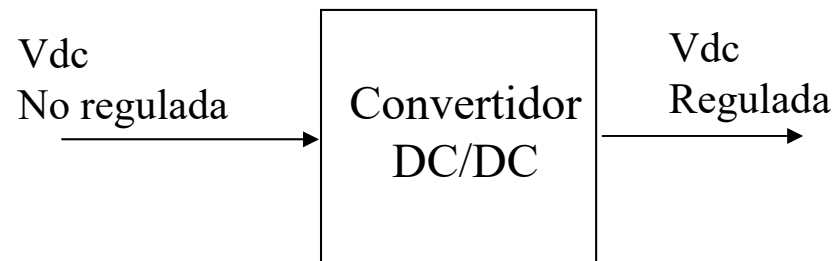
Bibliografía de consulta

- ELECTRÓNICA DE POTENCIA. Daniel W. Hart.
Editorial: Prentice-Hall. 2001.
✓ Páginas: 202-222

- DESIGN WITH OPERATIONAL AMPLIFIERS
AND ANALOG INTEGRATED CIRCUITS. (Fourth
Edition). Sergio Franco. Editorial: McGraw-Hill.
2015.
✓ Páginas: 566-607

- POWER ELECTRONICS: CONVERTERS,
APPLICATIONS AND DESIGN. (Third Edition).
Ned Mohan, Tore M. Undeland, William P. Robbins.
Editorial: John Wiley. 2003.
✓ Páginas: 161-184

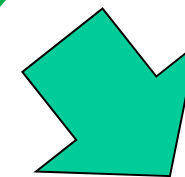
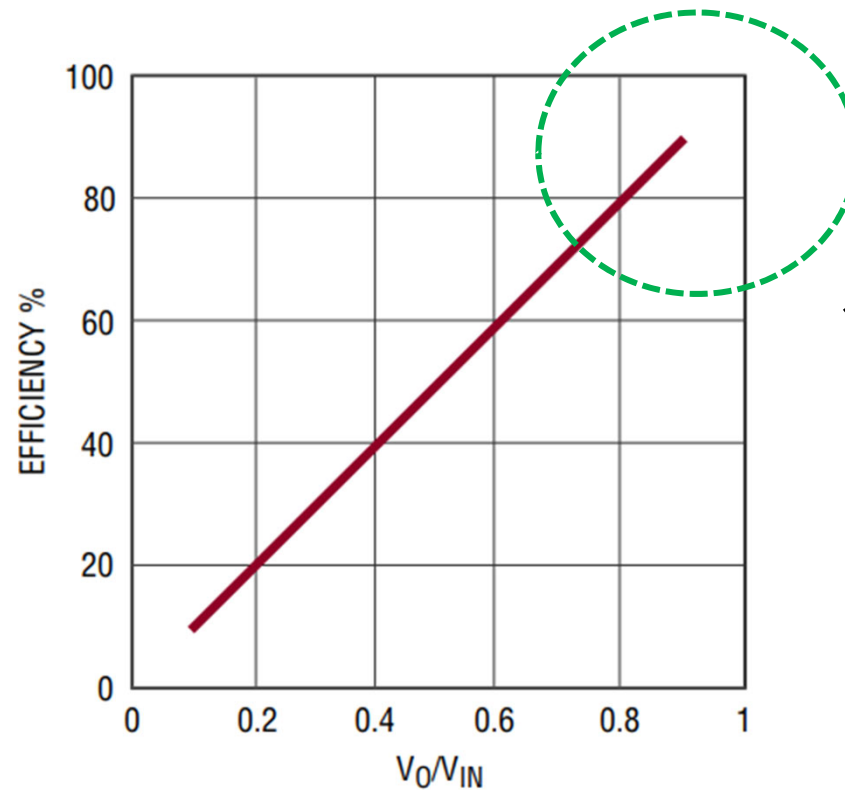
Convertidores conmutados DC/DC sin aislamiento



- Transistor en **conmutación**
- Elevado **rendimiento** (75 - 90%)
- **Topologías básicas:**
 - ✓ Convertidor **reductor** (buck, step down)
 - ✓ Convertidor **elevador** (boost, step up)
 - ✓ Convertidor **elevador/reductor** inversor (buck-boost)

Rendimiento en un regulador lineal

$$\eta_{LR} = \frac{P_{OUTPUT}}{P_{OUTPUT} + P_{LOSS}} = \frac{V_O \cdot I_O}{V_O \cdot I_O + (V_{IN} - V_O) \cdot I_O} = \frac{V_O}{V_{IN}}$$



**Límite:
Tensión de
Drop-out**

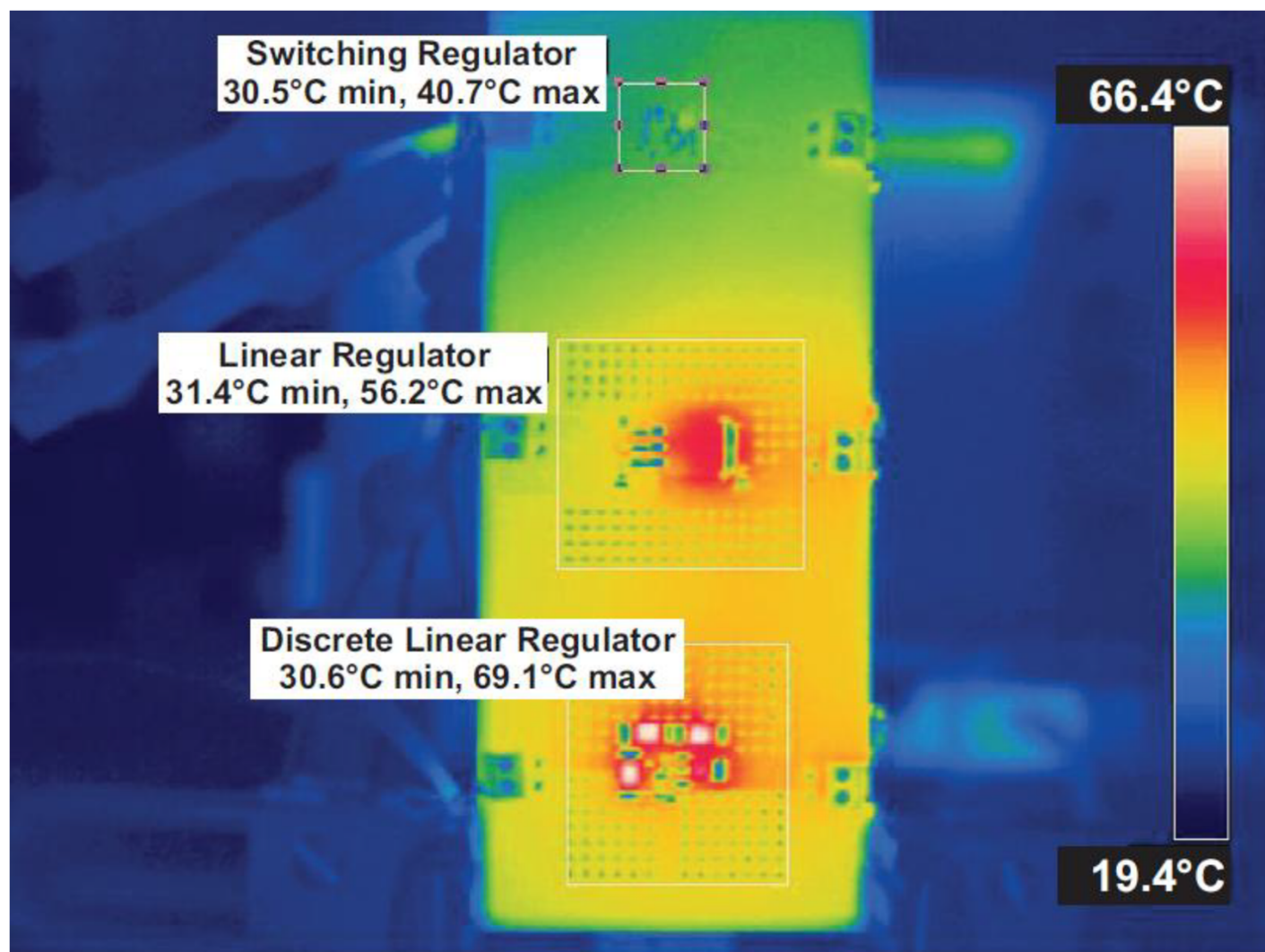


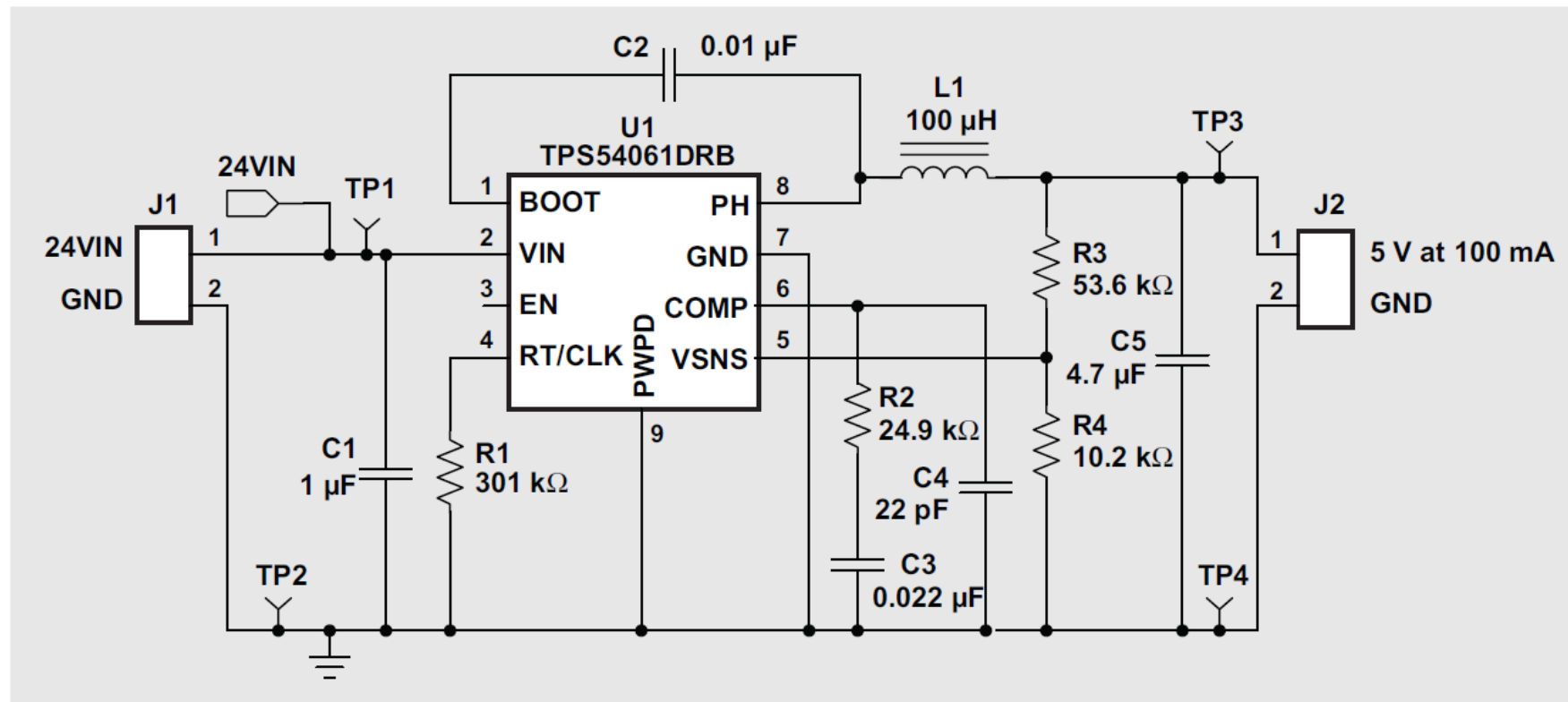
Figure 1. Switching (buck) converter with integrated MOSFETs

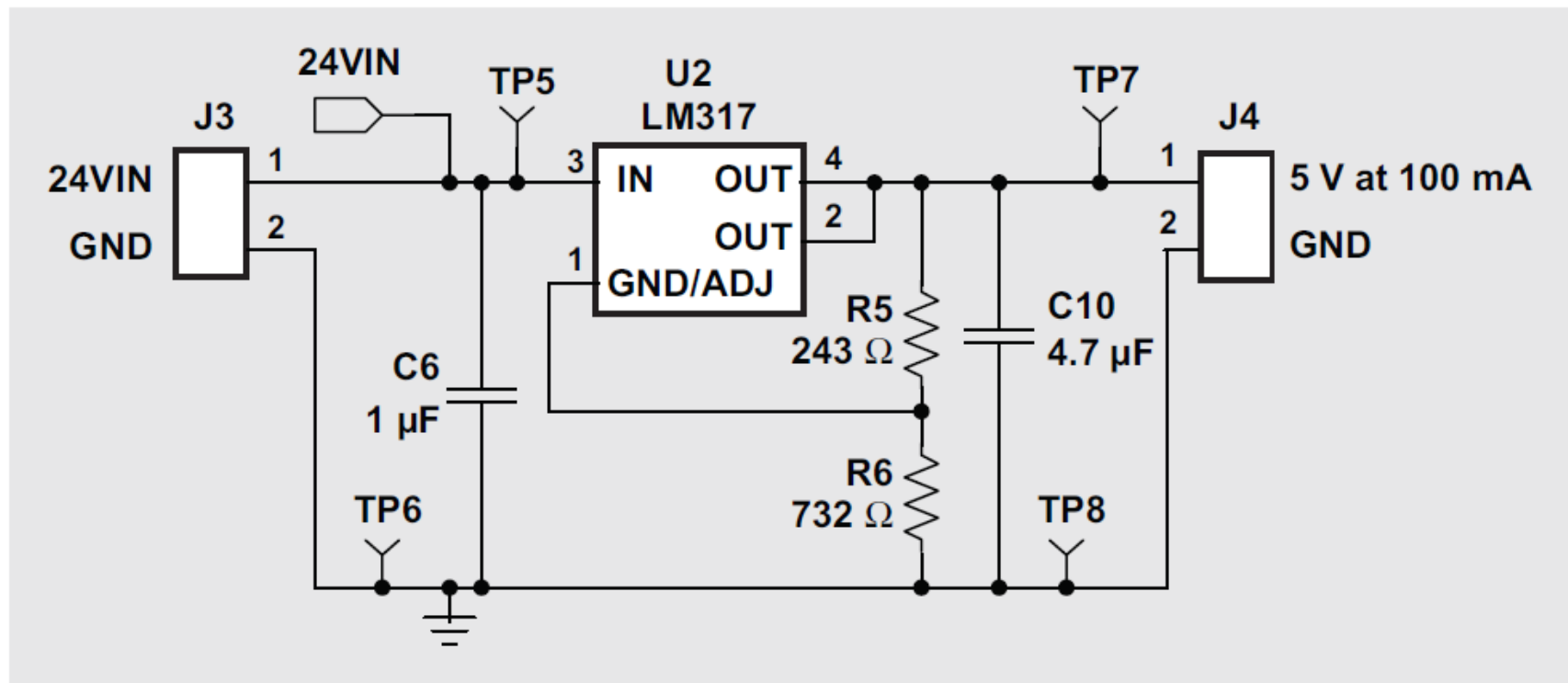
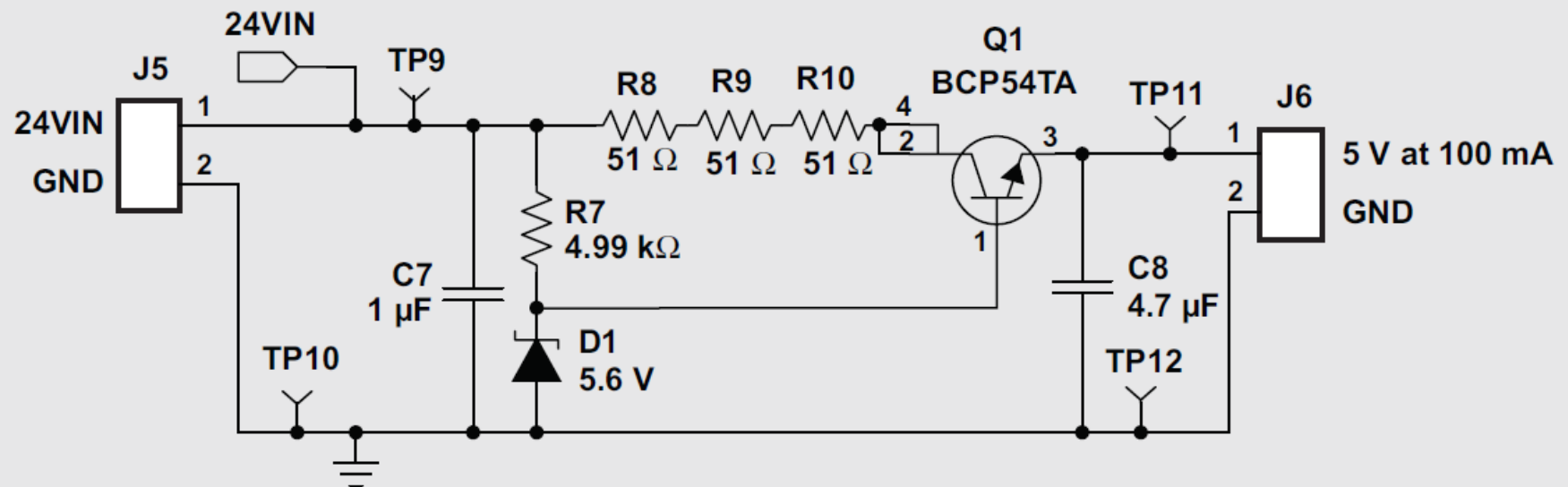
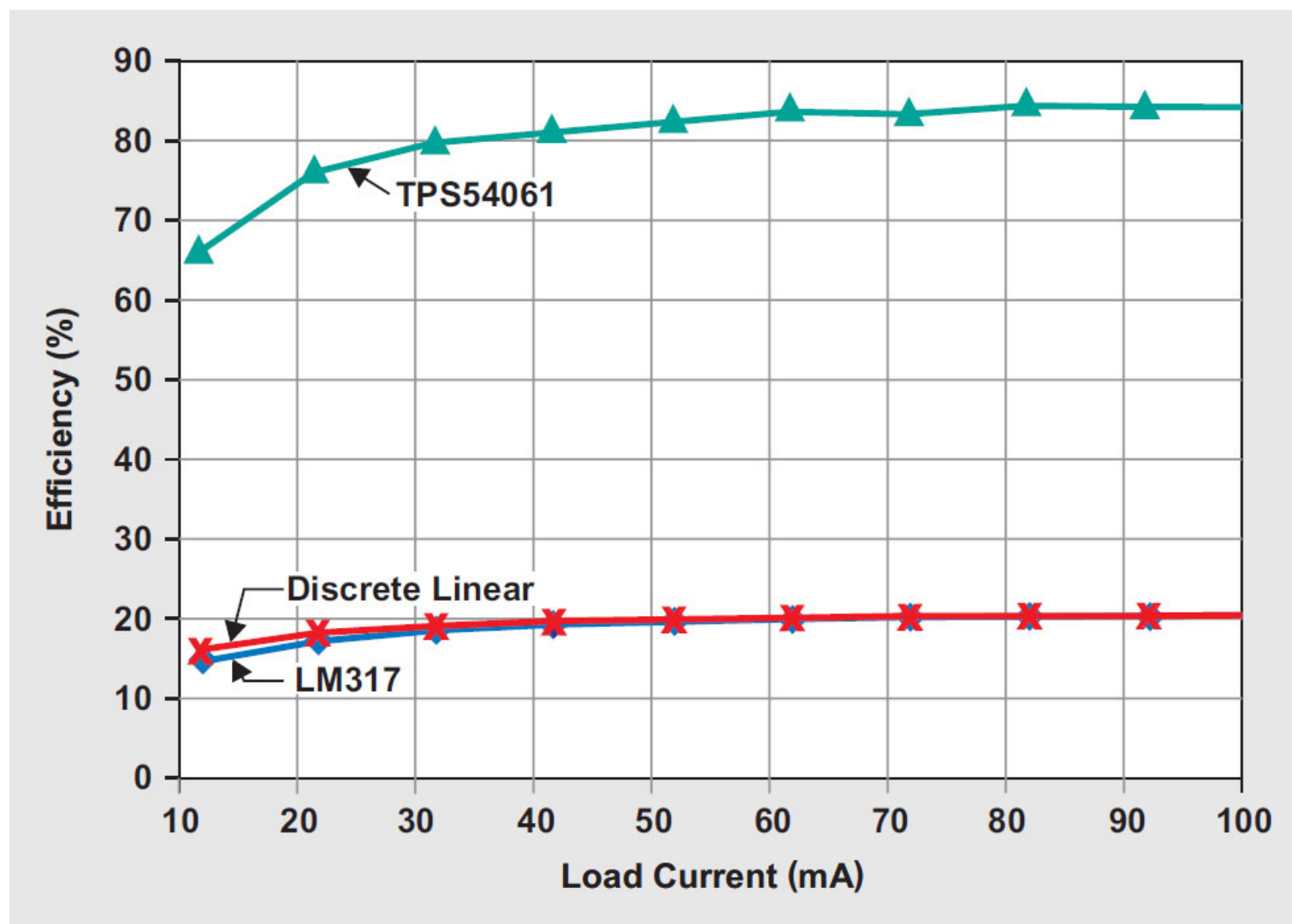
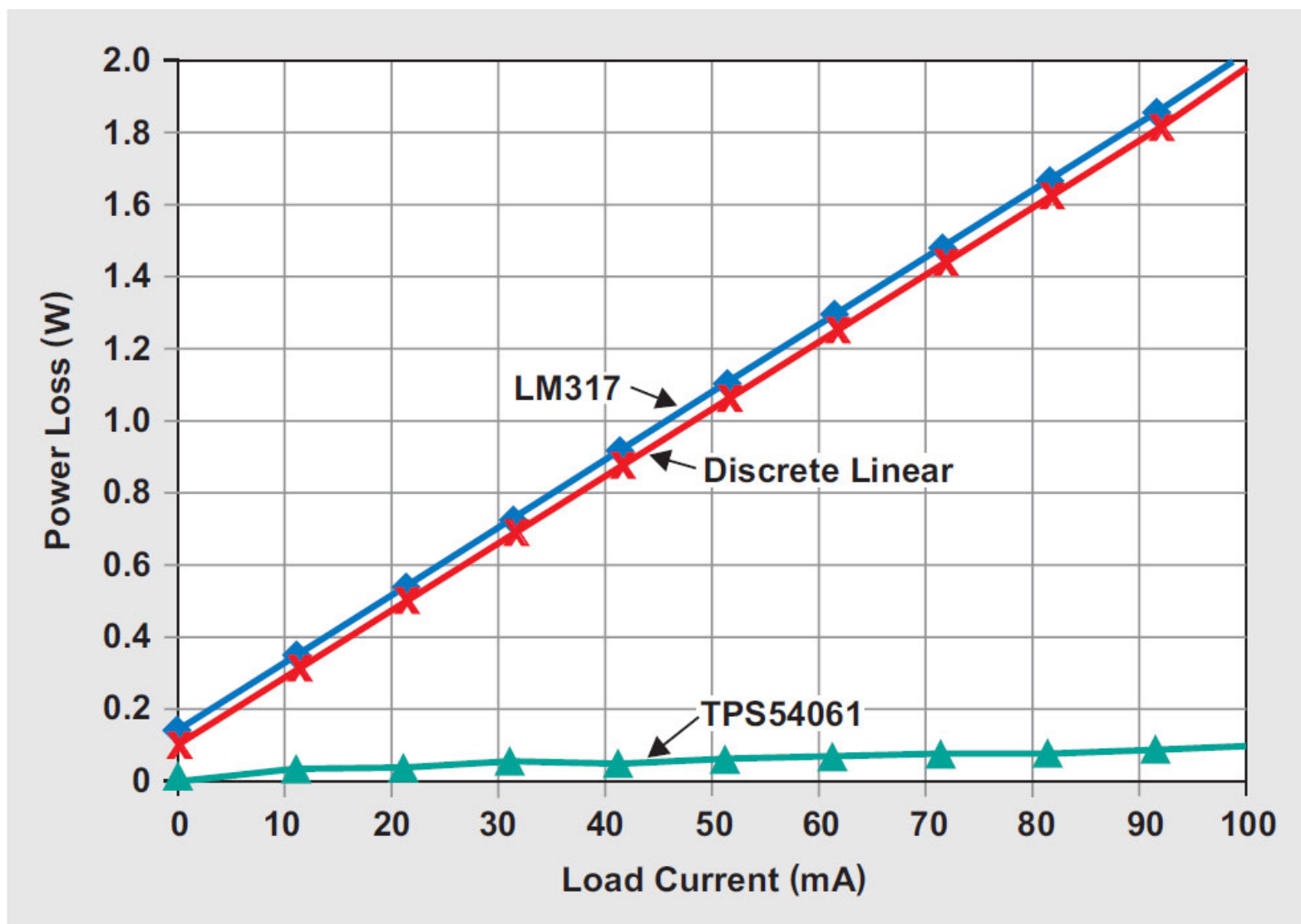
Figure 2. Integrated, wide-input-voltage linear regulator

Figure 3. Discrete linear regulator



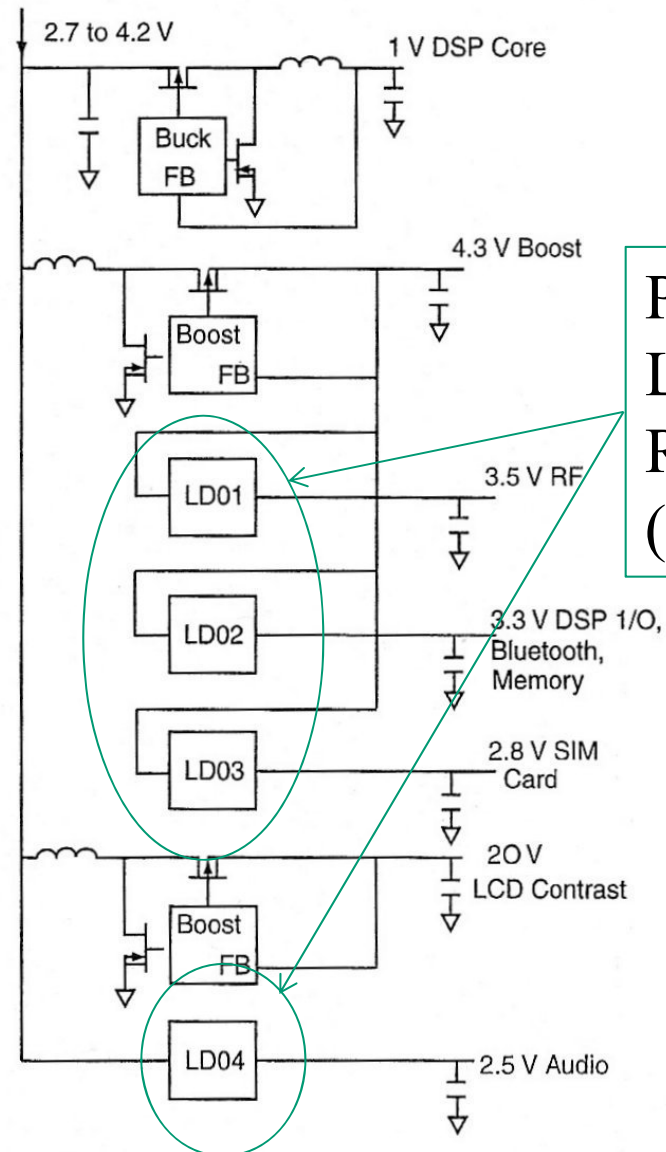


<https://www.ti.com/lit/slyt527>



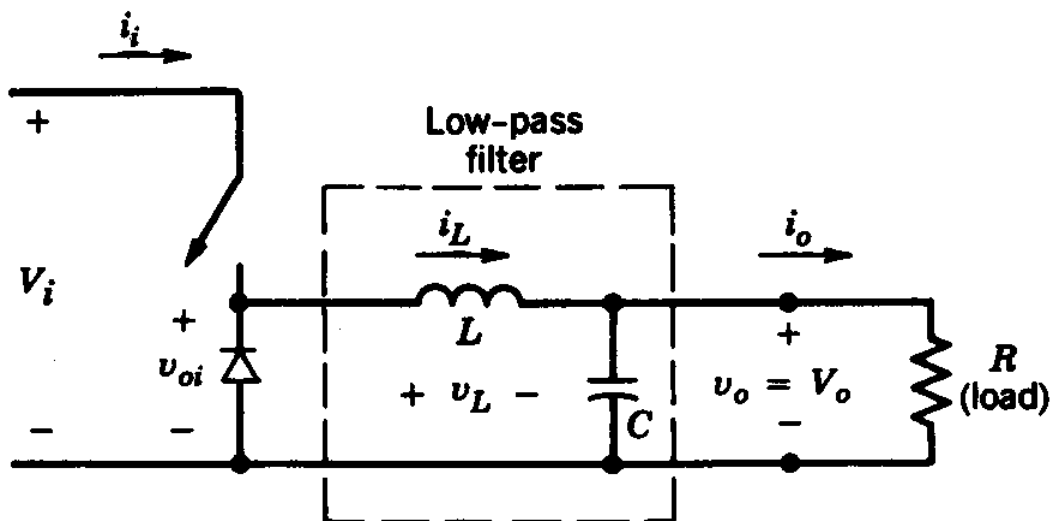
<https://www.ti.com/lit/slyt527>

Ejemplo: Alimentaciones en un teléfono celular

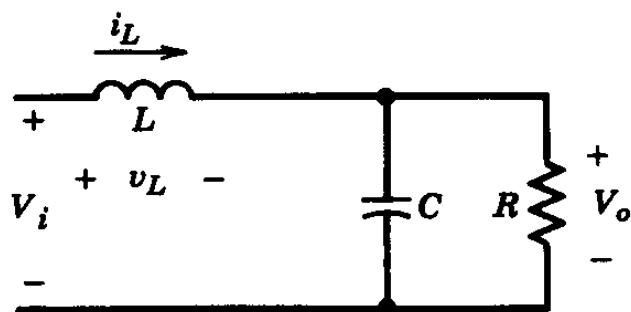


Post-regulación
Lineal mediante
Reguladores LDO
(Tema 3)

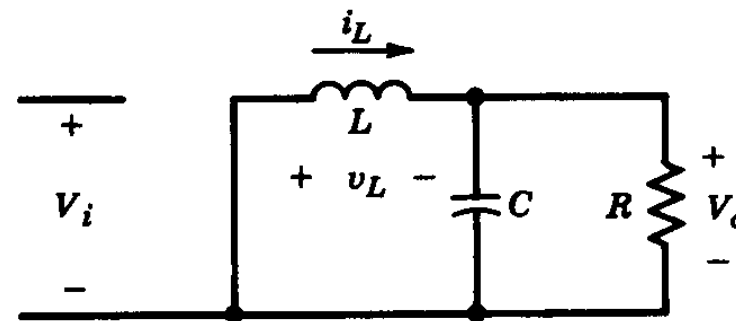
Convertidor reductor



Esquema

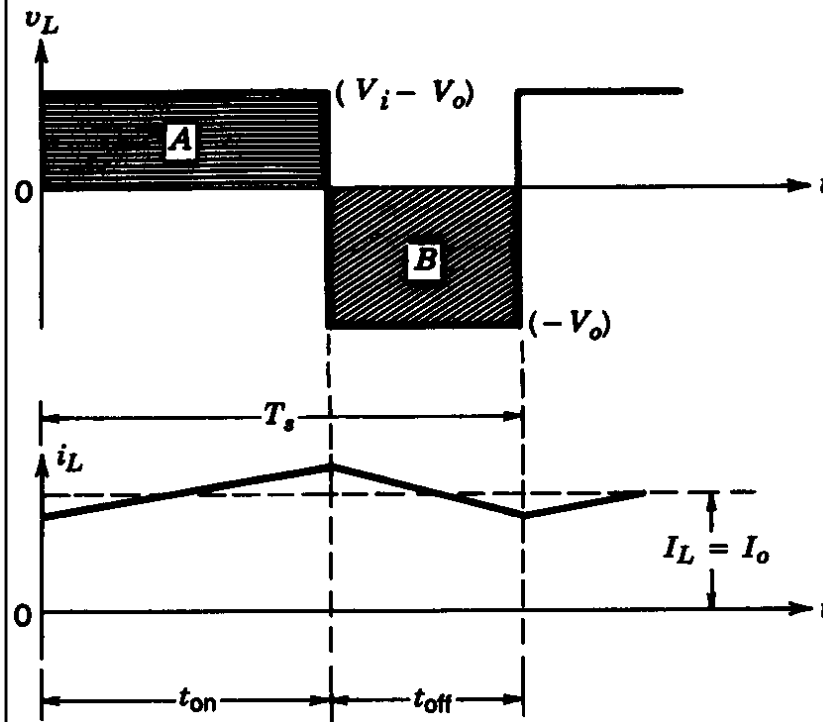


Circuito equivalente
transistor **on**



Circuito equivalente
transistor **off**

Convertidor reductor (modo continuo ideal)



Tensión y corriente en la bobina

Tensión en L

$$\int_0^{T_s} v_L(t) \cdot dt = \int_0^{t_{on}} v_L(t) \cdot dt + \int_{t_{on}}^{T_s} v_L(t) \cdot dt = 0$$

$$T_s = t_{on} + t_{off}$$

$$(V_i - V_o) \cdot t_{on} = V_o (T_s - t_{on})$$

$$\frac{V_o}{V_i} = \frac{t_{on}}{T_s} = D ; (D = \text{ciclo de trabajo})$$

Corriente en L

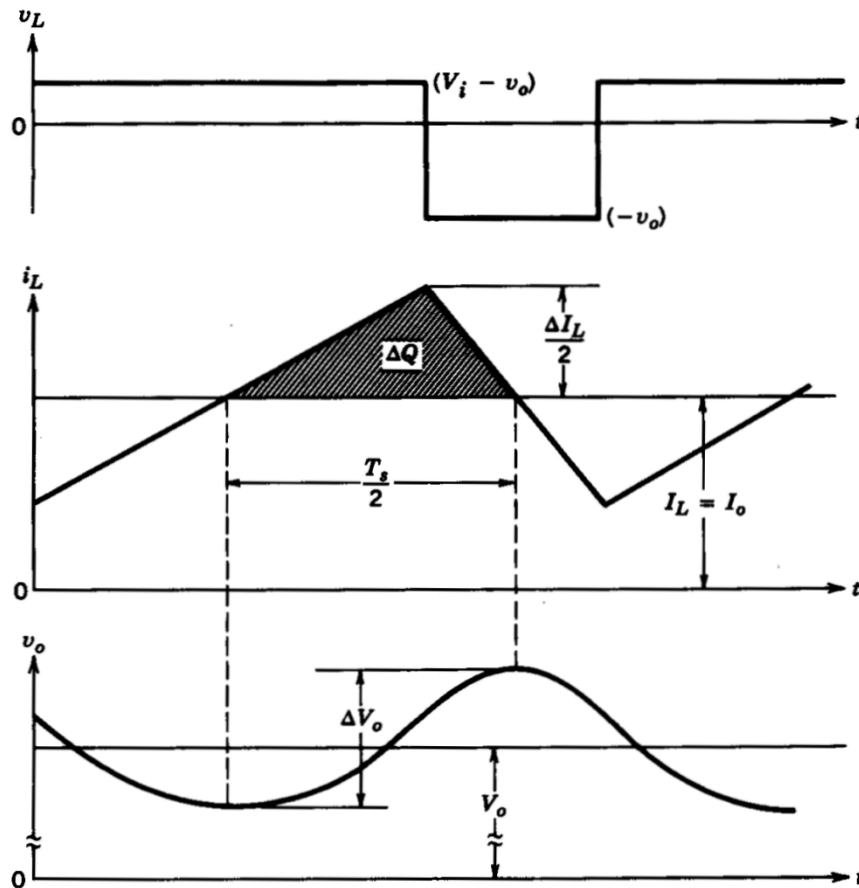
$$i_L(t) = \frac{1}{L} \int v_L(t) dt$$

$$\Delta I_L^+ = \frac{V_i - V_o}{L} t_{on}$$

$$-\Delta I_L^- = \frac{V_o}{L} t_{off}$$

$$\Delta I_L^+ = -\Delta I_L^- \Rightarrow V_o = V_i D \quad 14$$

Convertidor reductor (modo continuo ideal)



$$v_C(t) = \frac{1}{C} \int i_C(t) dt$$

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{1}{2} \frac{\Delta I_L}{2} \frac{T}{2} \frac{1}{C} = \frac{\Delta I_L T}{8C}$$

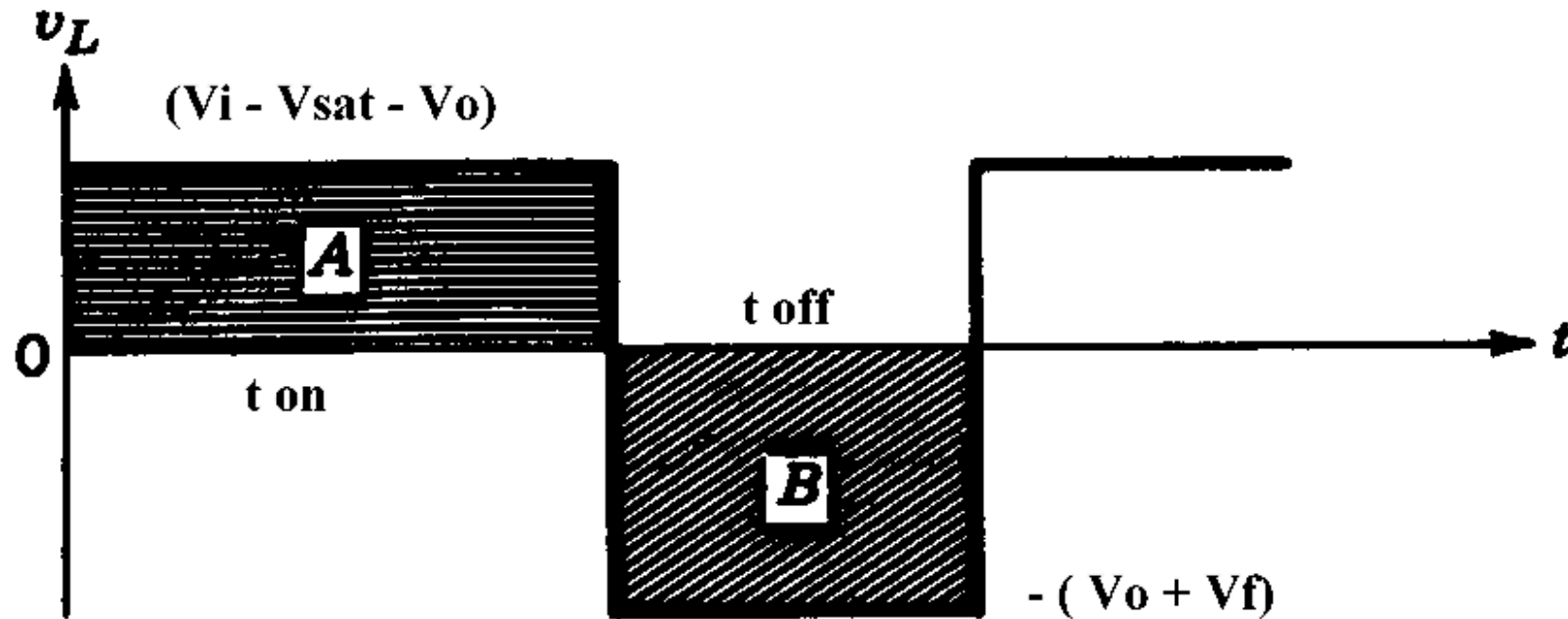
Rizado en la tensión de salida
(Capacidad ideal)

DISEÑO
EN MODO
CONTINUO
IDEAL

Diodo de conmutación	
V_{RRM}	$\geq V_i \cdot (MS) ; (MS=1,5 \text{ p. ej.})$
I_{FAV}	$\geq I_{OMAX} \cdot (1 - \delta)$
I_{FRM}	$\geq I_{OMAX} + \frac{(V_i - V_o)}{2 \cdot L} \cdot \delta \cdot T$
Transistor	
V_{CEO} BV_{DSS}	$\geq V_i \cdot (MS) ; (MS=1,5 \text{ p. ej.})$
I_{CDCMAX} I_{DDCMAX}	$\geq I_{OMAX} \cdot \delta$
I_{CM} I_{DM}	$\geq I_{OMAX} + \frac{V_o}{2 \cdot L} \cdot (1 - \delta) \cdot T$
Bobina (criterios)	
1 : Fijado modo de trabajo, L modo frontera	
2 : Fijar rizado de corriente, $\Delta i_L \leq 0,1 \cdot I_L$	
Condensador (ideal)	
Fijar rizado de tensión, $C = \frac{\Delta I_L \cdot T}{8 \cdot \Delta V_o}$	

CONVERTIDOR
REDUCTOR

Convertidor reductor (modo continuo no ideal)

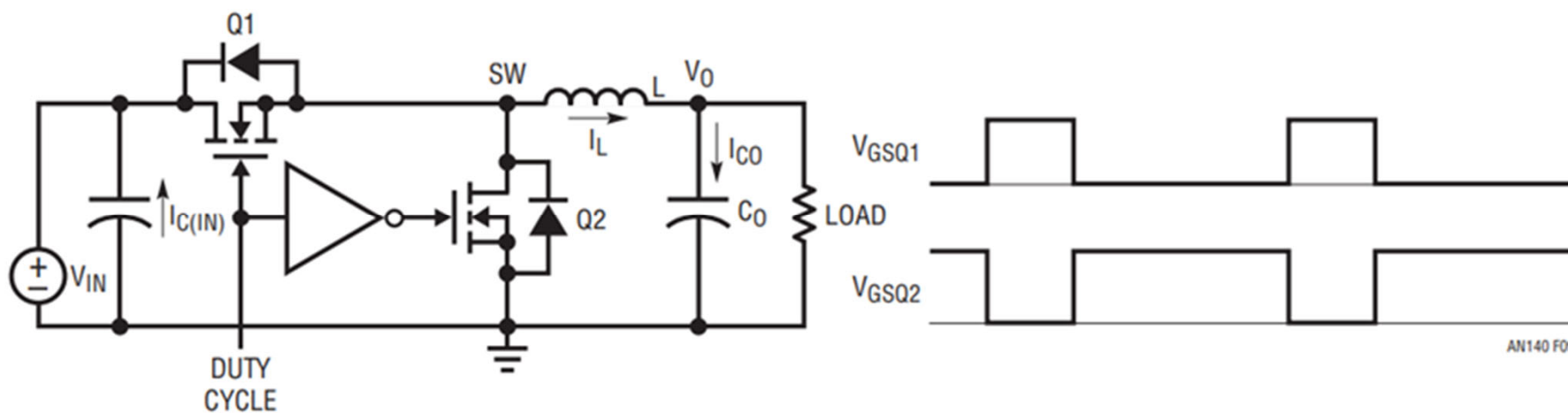


Considerando las caídas de tensión en transistor y diodo:

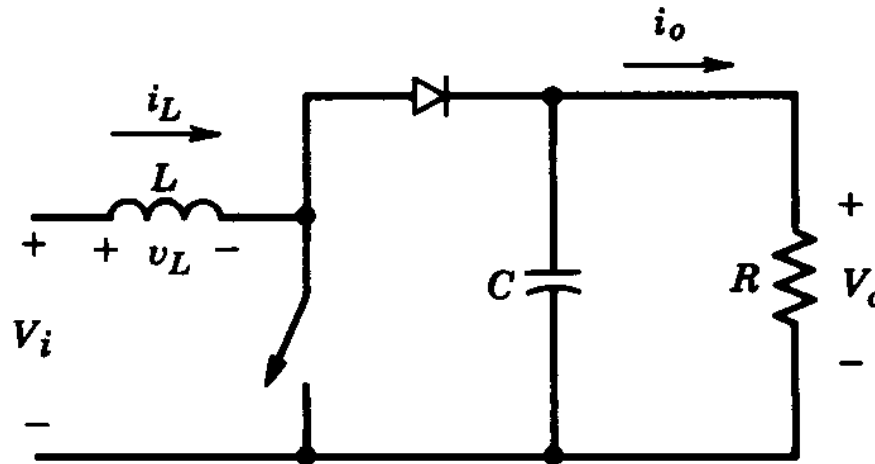
$$(V_i - V_{sat} - V_o) \cdot t_{on} = (V_o + V_f) \cdot t_{off}$$

$$V_o = (V_i - V_{sat})D - V_f(1 - D)$$

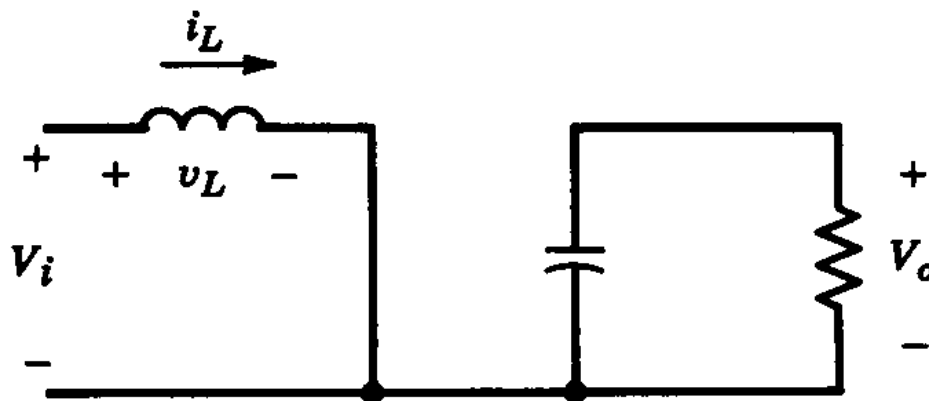
DC/DC reductor síncrono



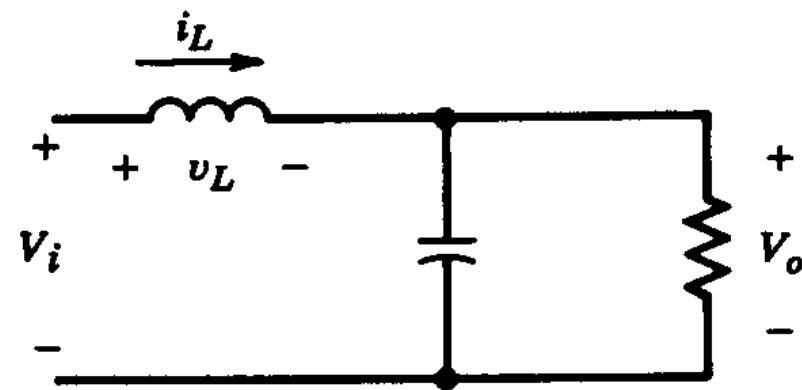
Convertidor elevador



Esquema

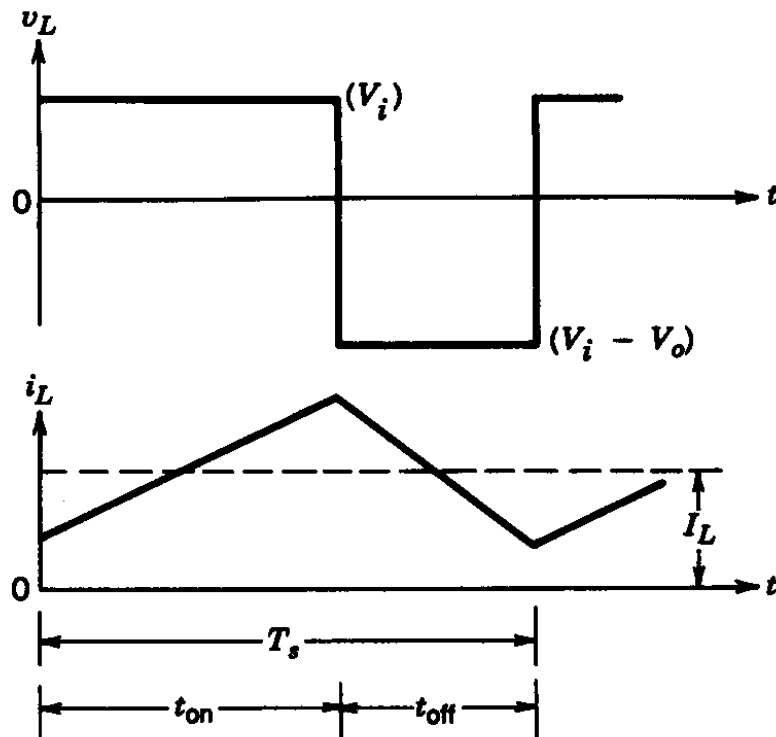


Circuito equivalente
transistor **on**



Circuito equivalente
transistor **off**

Convertidor elevador (modo continuo ideal)



Tensión y corriente en la bobina

Tensión en la bobina

$$\int_0^{T_s} v_L(t) \cdot dt = \int_0^{t_{on}} v_L(t) \cdot dt + \int_{t_{on}}^{T_s} v_L(t) \cdot dt = 0$$

$$T_s = t_{on} + t_{off}$$

$$V_i \cdot t_{on} = -(V_i - V_o) t_{off}$$

$$\frac{V_o}{V_i} = \frac{T_s}{t_{off}} = \frac{1}{1-D} \Rightarrow V_o = \frac{V_i}{1-D}$$

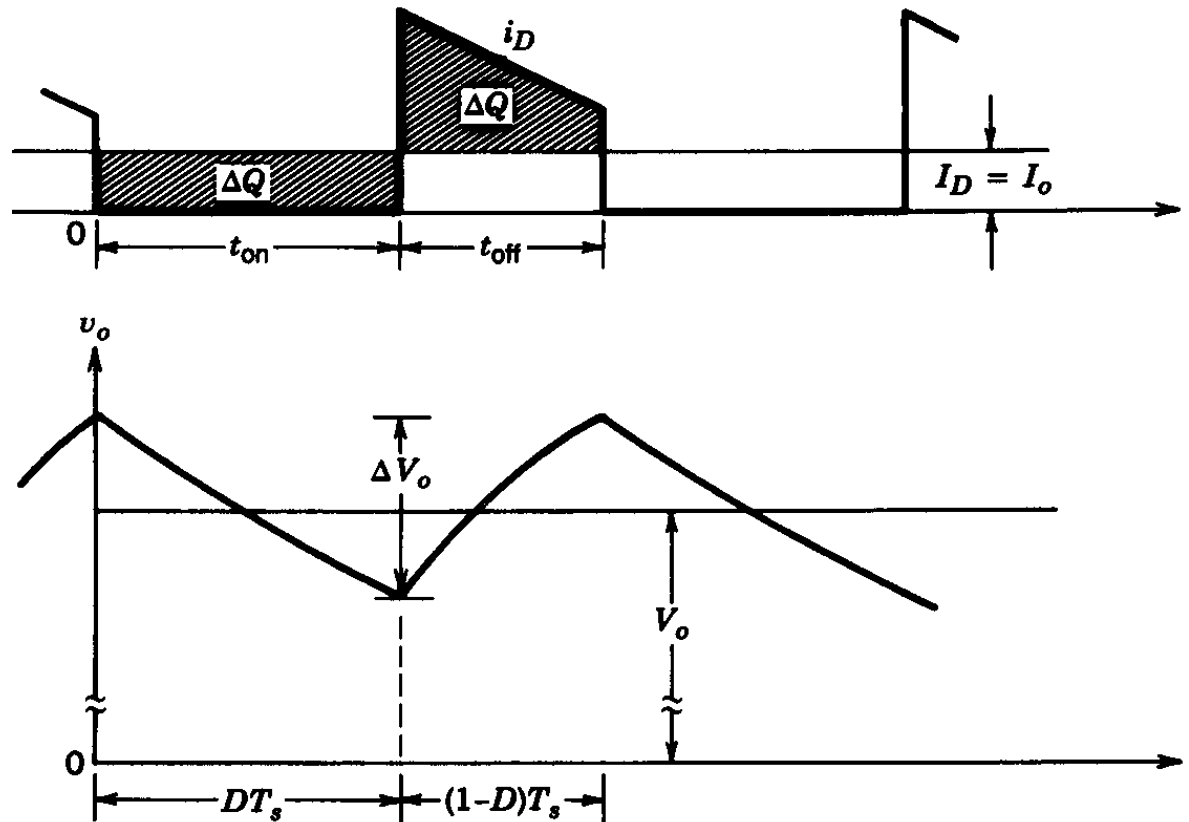
Corriente en la bobina

$$\Delta I_L^+ = \frac{V_i}{L} t_{on}$$

$$-\Delta I_L^- = \frac{V_o - V_i}{L} t_{off}$$

$$\Delta I_L^+ = -\Delta I_L^- \Rightarrow V_o = \frac{V_i}{1-D}$$

Convertidor elevador (modo continuo ideal)



Si :

$$I_o = cte.$$

$$\Delta Q = I_o t_{on}$$

$$\Delta V_o = \frac{1}{C} I_o t_{on}$$

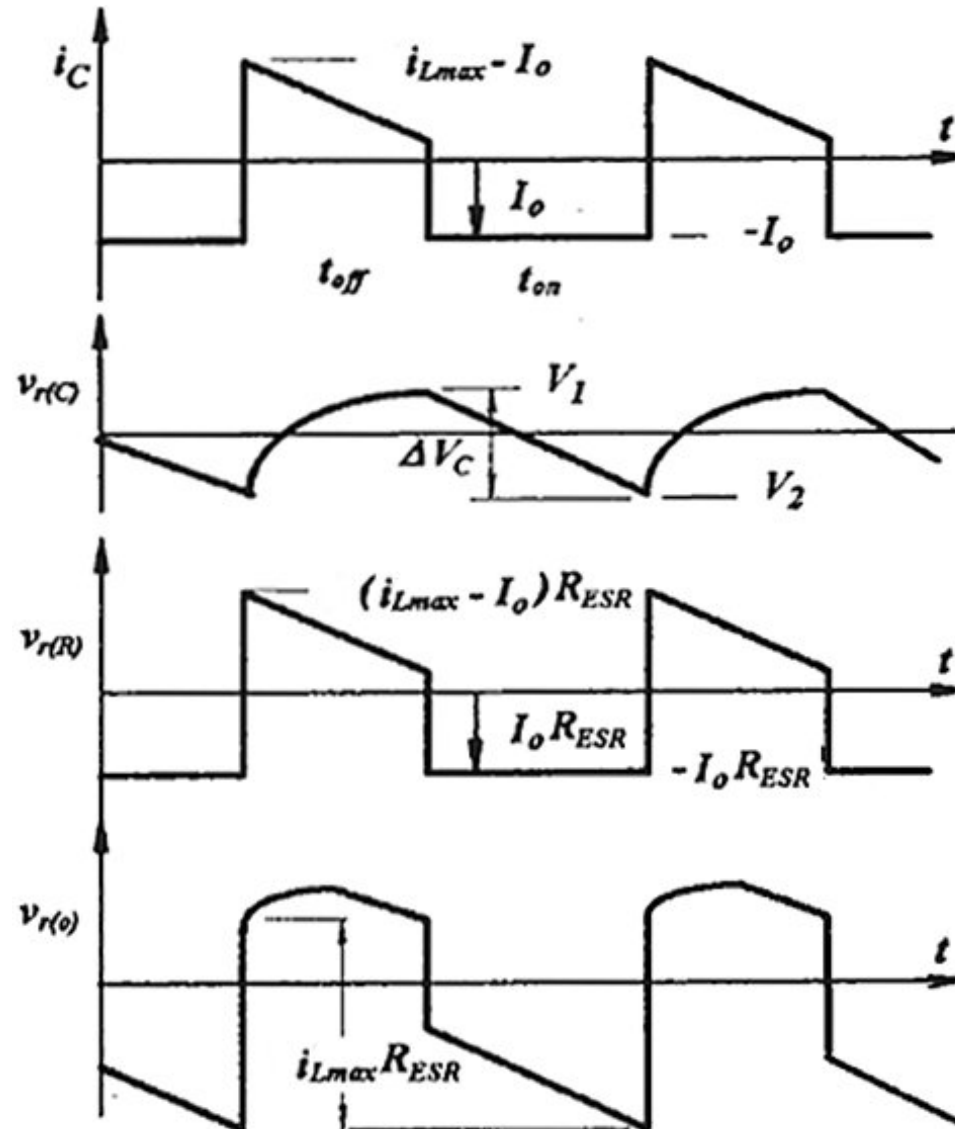
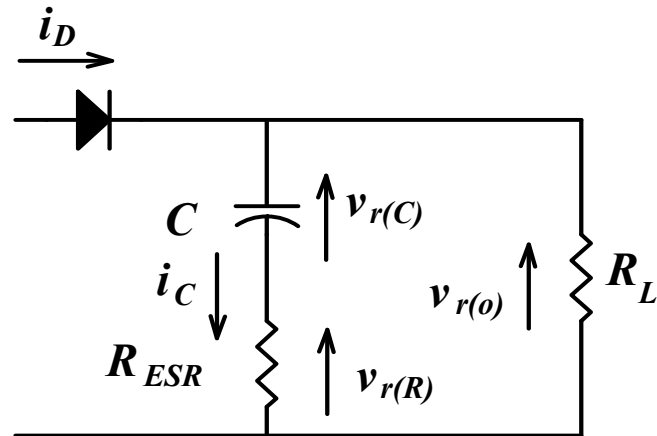
Tensión de rizado
(C ideal)

DISEÑO
EN MODO
CONTINUO
IDEAL

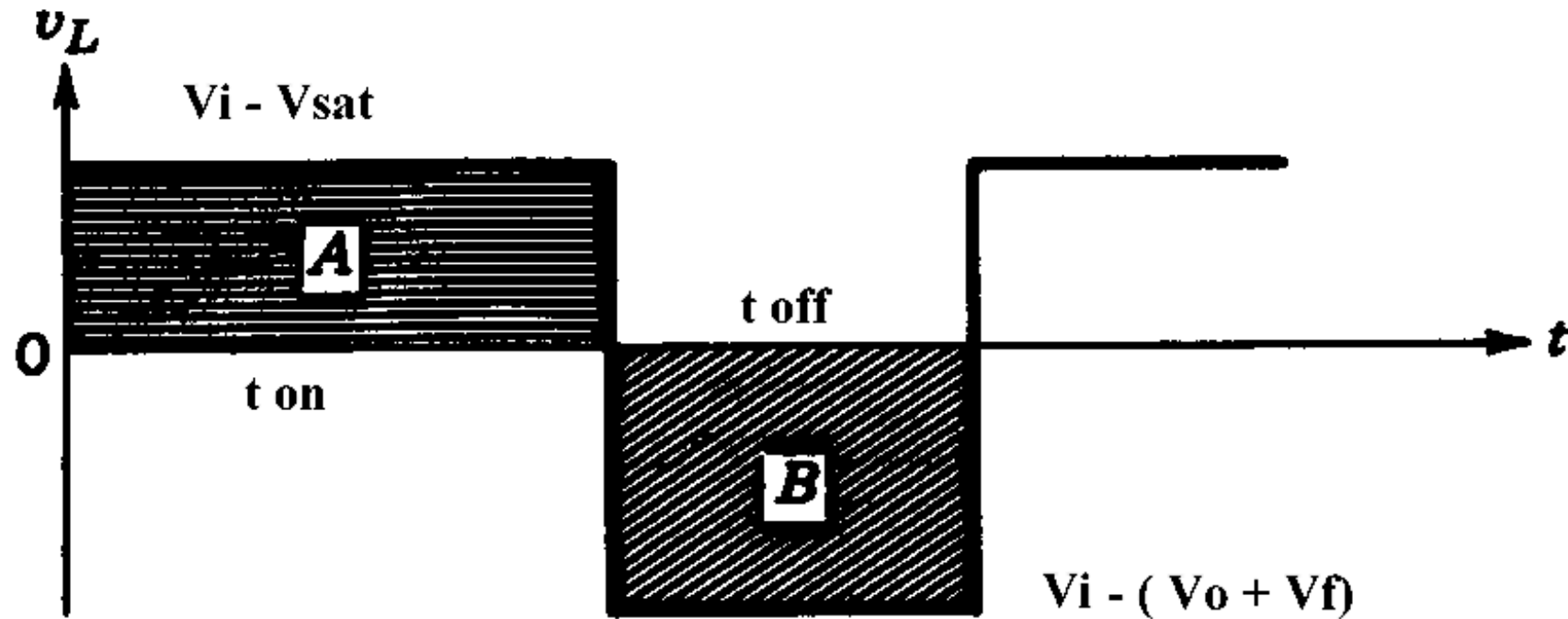
Diodo de conmutación	
V_{RRM}	$\geq V_o \cdot (MS) ; (MS=1,5 \text{ p. ej.})$
I_{FAV}	$\geq I_{OMAX}$
I_{FRM}	$\geq \frac{I_{OMAX}}{(1-\delta)} + \frac{V_i}{2 \cdot L} \cdot \delta \cdot T$
Transistor	
V_{CEO} BV_{DSS}	$\geq V_o \cdot (MS) ; (MS=1,5 \text{ p. ej.})$
$I_{CDCMÁX}$ $I_{DDCMÁX}$	$\geq \frac{I_{OMAX}}{(1-\delta)} \cdot \delta$
I_{CM} I_{DM}	$\geq \frac{I_{OMAX}}{(1-\delta)} + \frac{(V_o - V_i)}{2 \cdot L} \cdot (1-\delta) \cdot T$
Bobina (criterios)	
1 : Fijado modo de trabajo, L modo frontera	
2 : Fijar rizado de corriente, $\Delta i_L \leq 0,1 \cdot I_L$	
Condensador (ideal)	
Fijar rizado de tensión, $C = \frac{I_{OMAX} \cdot \delta \cdot T}{\Delta V_o}$	

CONVERTIDOR
ELEVADOR

Efecto de la ESR del condensador



Convertidor elevador (modo continuo no ideal)

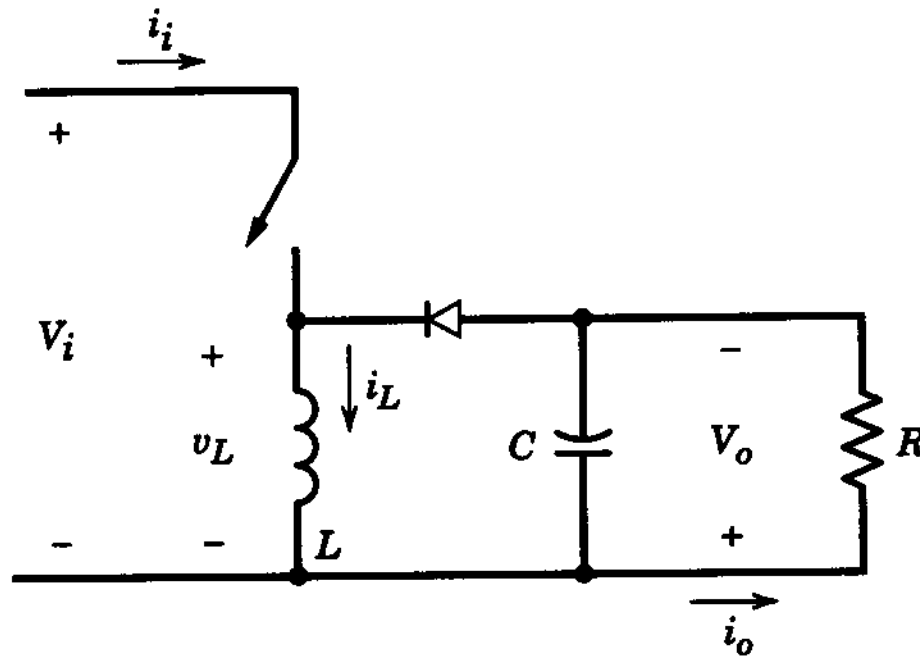


Considerando las caídas de tensión en transistor y diodo:

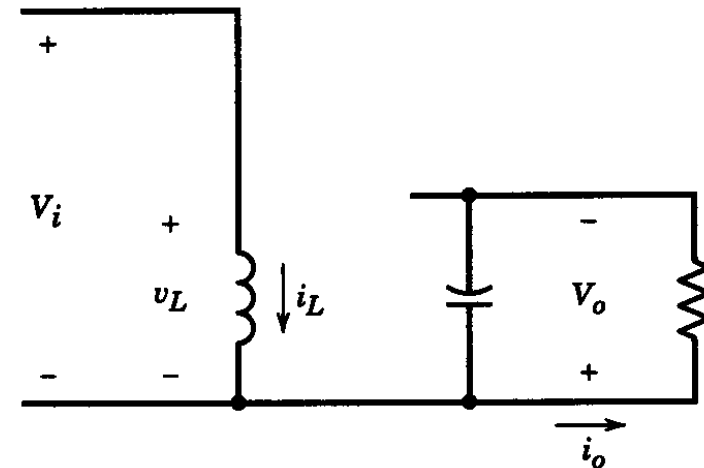
$$(V_i - V_{sat}) \bullet t_{on} = (V_o + V_f - V_i) \bullet t_{off}$$

$$V_o = \frac{V_i}{(1 - D)} - V_{sat} \frac{D}{(1 - D)} - V_f$$

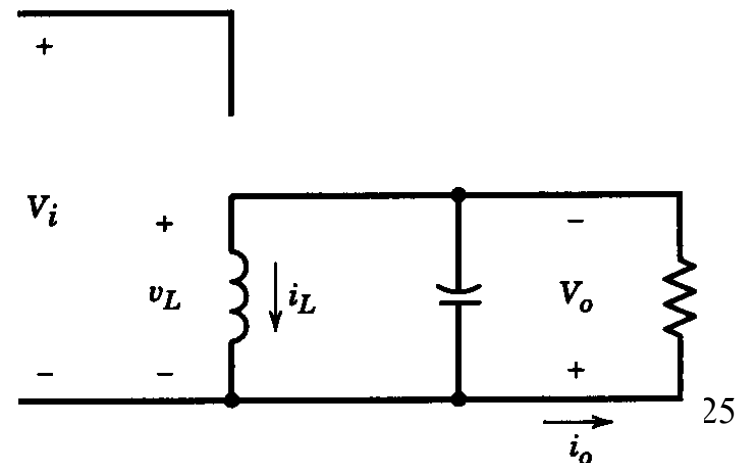
Convertidor elevador/reductor (inversor)



Esquema ¡ojo! V_o definida negativa respecto al nudo común

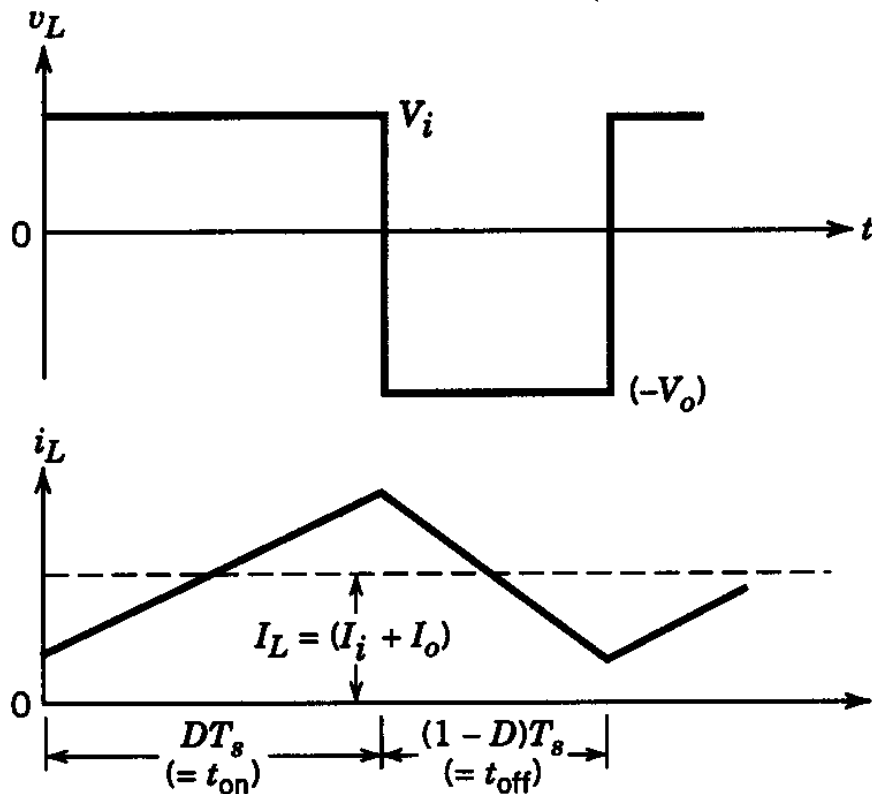


Circuito equivalente transistor **on**



Circuito equivalente transistor **off**

Convertidor elevador/reductor (modo continuo ideal)



Tensión y corriente en la bobina

Tensión en la bobina

$$V_i t_{on} + (-V_o) t_{off} = 0$$

$$\frac{V_o}{V_i} = \frac{D}{1-D} \Rightarrow V_o = V_i \frac{D}{1-D}$$

Corriente en la bobina

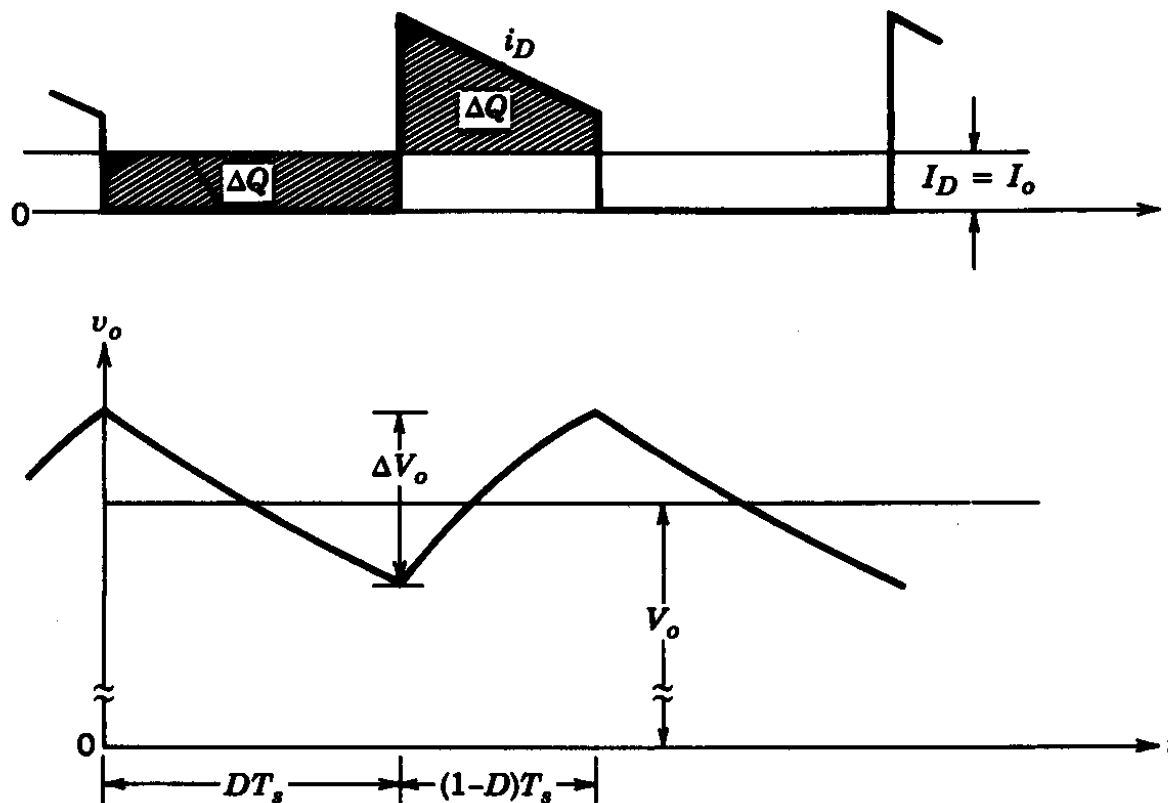
$$\Delta I_L^+ = \frac{V_i}{L} t_{on}$$

$$-\Delta I_L^- = \frac{V_o}{L} t_{off}$$

$$\Delta I_L^+ = -\Delta I_L^- \Rightarrow V_o = V_i \frac{D}{1-D}$$

¡ojo! V_o definida negativa respecto al nudo común

Convertidor elevador/reductor (modo continuo ideal)



Si :

$$I_o = cte.$$

$$\Delta Q = I_o t_{on}$$

$$\Delta V_o = \frac{1}{C} I_o t_{on}$$

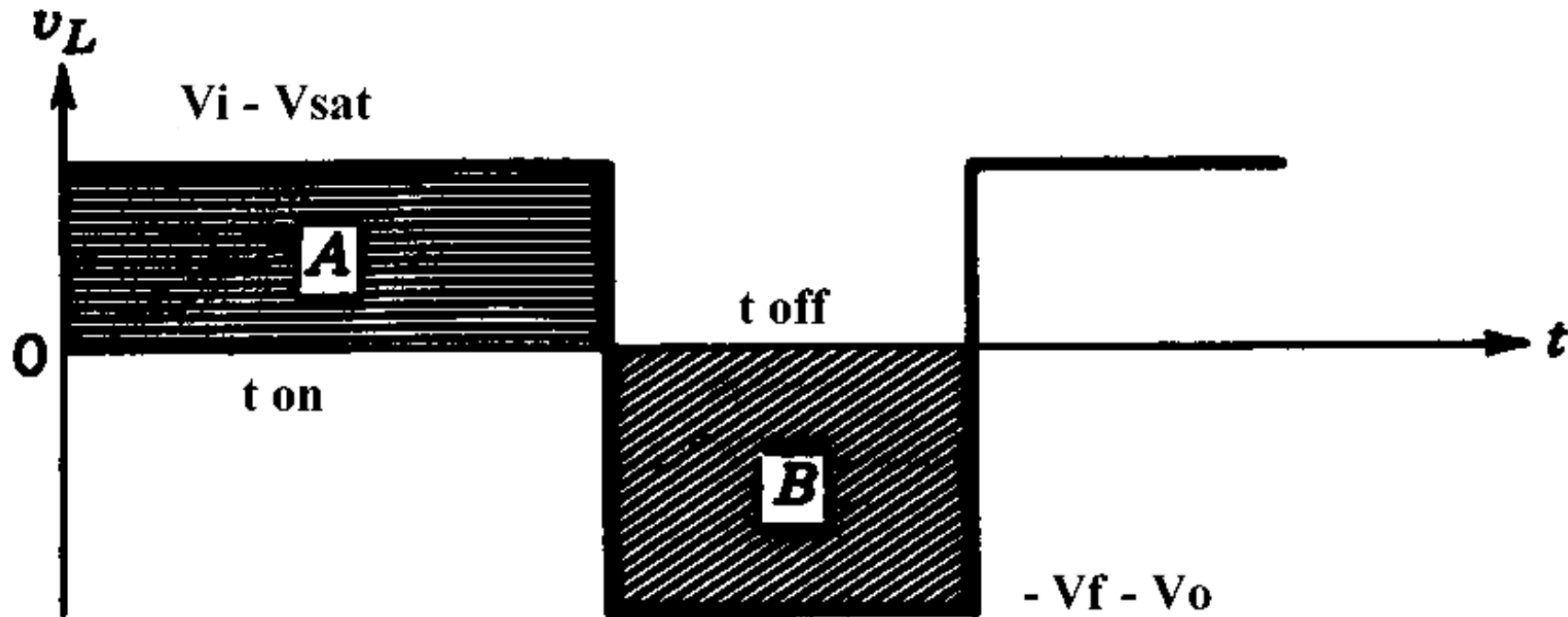
Tensión de rizado (C ideal)

DISEÑO
EN MODO
CONTINUO
IDEAL

Diodo de conmutación	
V_{RRM}	$\geq (V_i + V_o) \cdot (MS) ; (MS=1,5 \text{ p. ej.})$
I_{FAV}	$\geq I_{OMAX}$
I_{FRM}	$\geq \frac{I_{OMAX}}{(1-\delta)} + \frac{V_i}{2 \cdot L} \cdot \delta \cdot T$
Transistor	
V_{CEO} BV_{DSS}	$\geq (V_o + V_i) \cdot (MS) ; (MS=1,5 \text{ p. ej.})$
I_{CDCMAX} I_{DDCMAX}	$\geq \frac{I_{OMAX}}{(1-\delta)} \cdot \delta$
I_{CM} I_{DM}	$\geq \frac{I_{OMAX}}{(1-\delta)} + \frac{ V_o }{2 \cdot L} \cdot (1-\delta) \cdot T$
Bobina (criterios)	
1 : Fijado modo de trabajo, L modo frontera	
2 : Fijar rizado de corriente, $\Delta i_L \leq 0,1 \cdot I_L$	
Condensador (ideal)	
Fijar rizado de tensión, $C = \frac{I_{OMAX} \cdot \delta \cdot T}{\Delta V_o}$	

CONVERTIDOR
ELEVADOR/
REDUCTOR

Convertidor elevador/reductor (modo continuo no ideal)

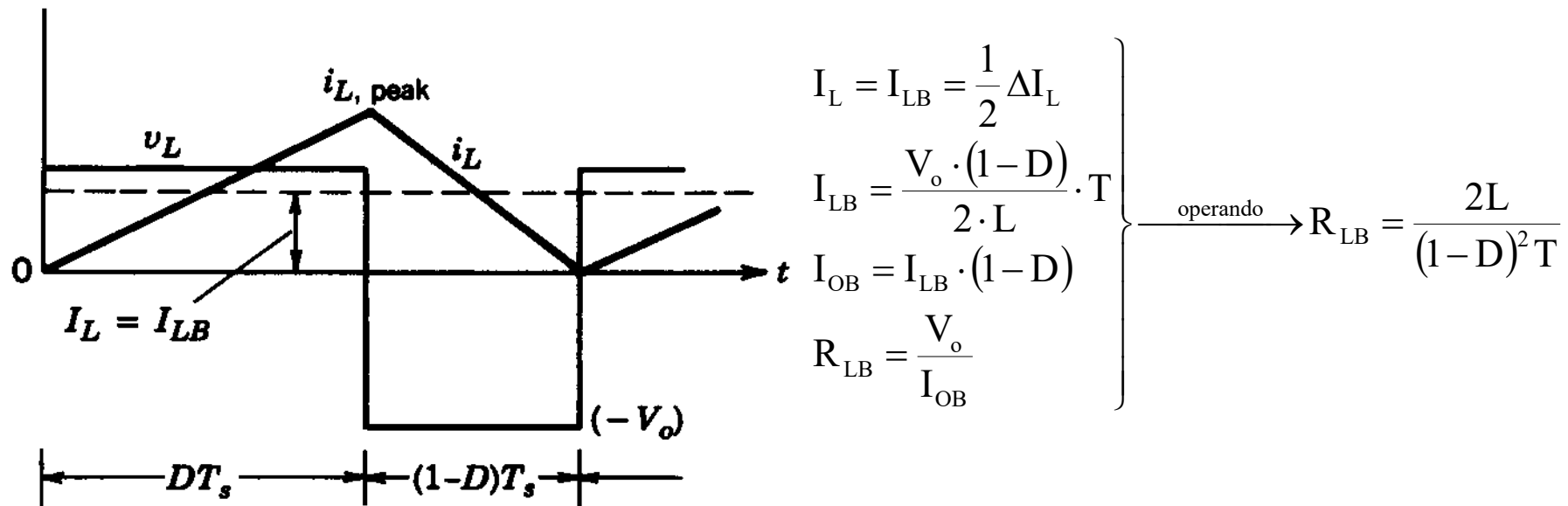


Considerando las caídas de tensión en transistor y diodo:

$$(V_i - V_{sat}) \bullet t_{on} = (V_o + V_f) \bullet t_{off}$$

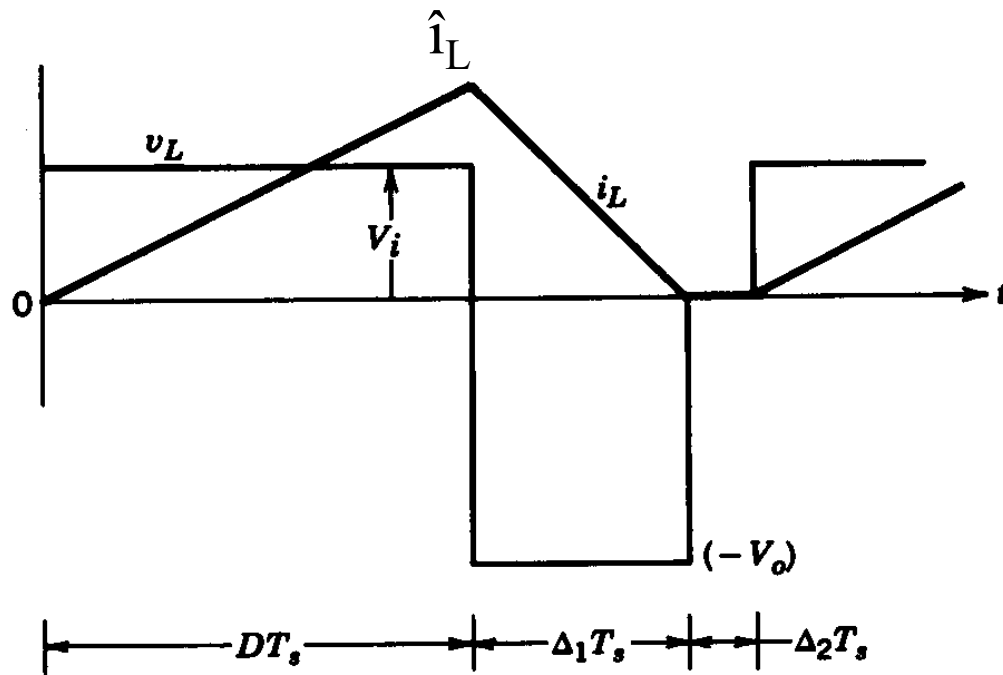
$$V_o = (V_i - V_{sat}) \frac{D}{(1 - D)} - V_f$$

Convertidor elevador/reductor (límite del modo continuo)



Modo frontera

Convertidor elevador/reductor (modo discontinuo ideal)



$$W_L = \frac{1}{2} \hat{i}_L^2 L = \frac{1}{2} \left(\frac{V_i}{L} t_{on} \right)^2 L = P_o T$$

$$P_o = \frac{V_o^2}{R_L}$$

$$V_o = V_i D \sqrt{\frac{TR_L}{2L}}$$

En este convertidor, en el modo discontinuo, la energía almacenada en la inductancia, se cede totalmente a la salida en cada ciclo de conmutación.

Modo **discontinuo ideal**

Convertidor	Ecuación (DCM ideal)
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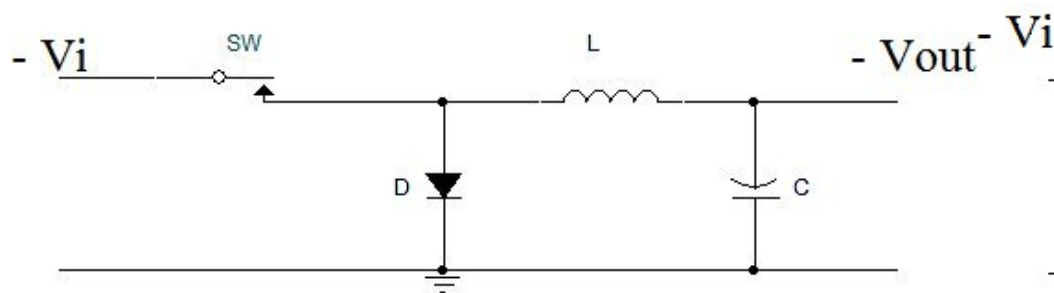
Reductor	$V_o = V_i \frac{D^2}{D^2 + \frac{2 \cdot L \cdot I_o}{T \cdot V_i}}$
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Elevador	$V_o = V_i \cdot \left(1 + \frac{V_i \cdot D^2 \cdot T}{2 \cdot L \cdot I_o} \right)$
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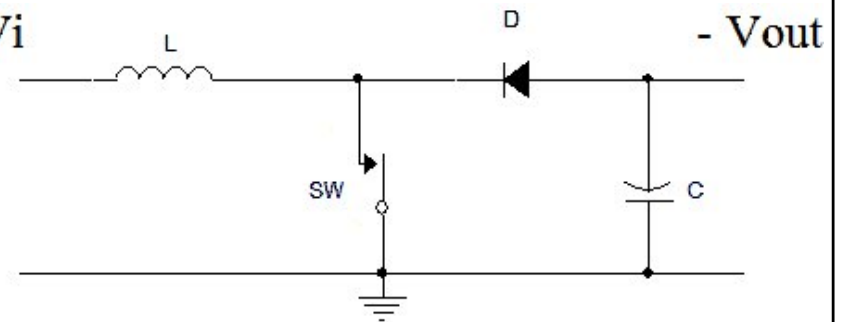
Elevador/ Reductor	$V_o = V_i \cdot D \sqrt{\frac{R_L \cdot T}{2 \cdot L}}$
--------------------	--

Convertidores DC/DC para tensión de entrada negativa.

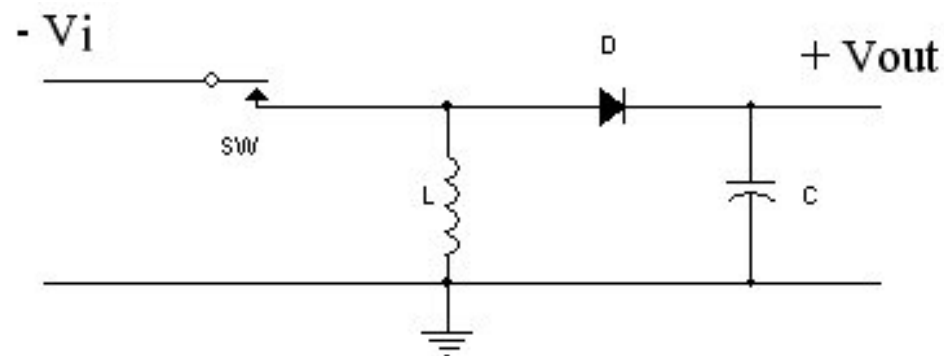
Convertidor reductor



Convertidor elevador



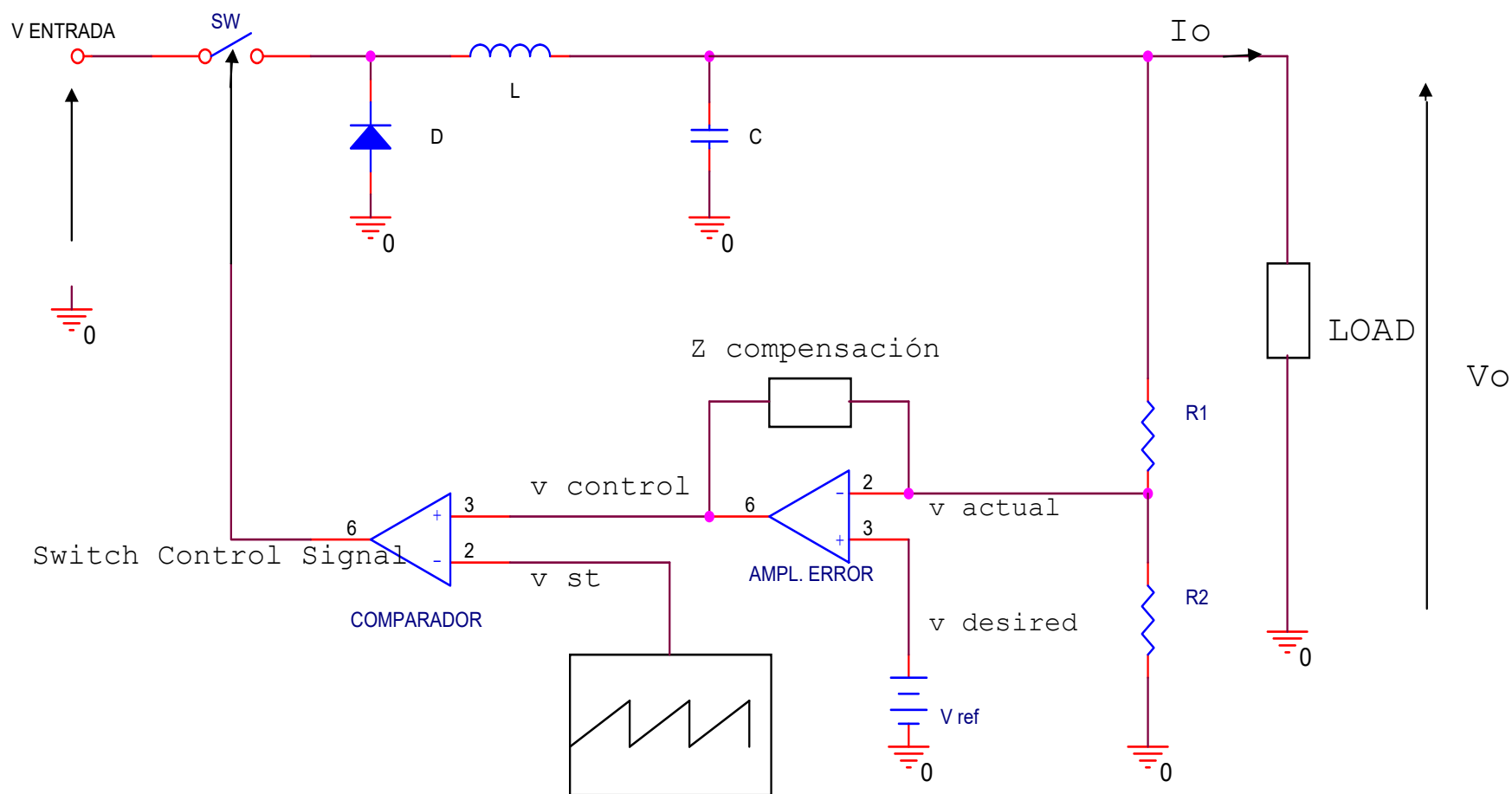
Convertidor elevador/reductor



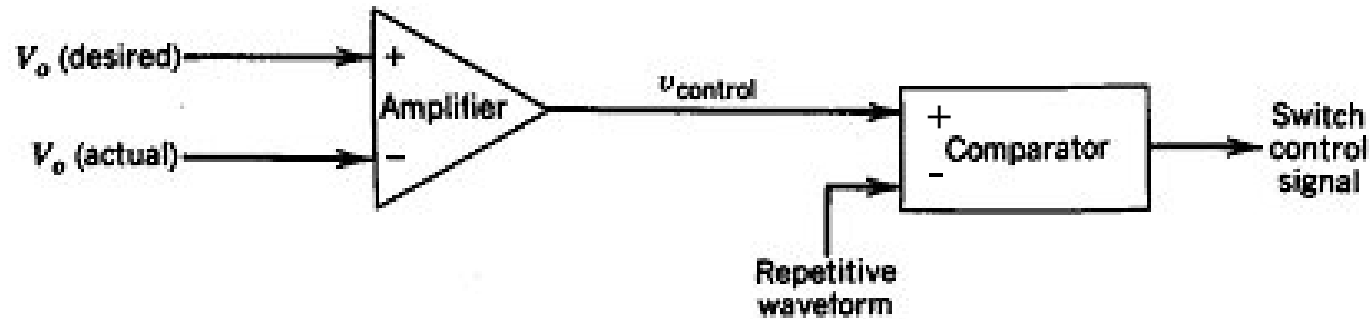
Control de los Convertidores DC/DC

- Control PWM (*Pulse -Width Modulation*)
 - Frecuencia de conmutación fija (de 10 kHz a 1MHz)
 - Elección de la frecuencia de conmutación. Compromiso: f grande $\rightarrow$ reducción tamaño L y C, pero mayores pérdidas de conmutación y generación de EMI.
 - Ciclo de trabajo ajustable mediante ton
 - Tipos: PWM modo tensión, PWM Feed-forward, PWM modo corriente (Si $D > 50\%$ $\rightarrow$ precisa slope compensation).
- Control PSM (*Pulse-Skip Modulation*) / PFM (*Pulse-Frequency Modulation*)
 - t_{on} o t_{off} fijos
 - “Desaparecen” ciclos de conmutación
 - Ciclo de trabajo ajustable mediante frecuencia de conmutación variable
 - Problema: EMI barre mayor espectro (bajas frecuencias).
 - Ciclo de trabajo pequeño $>$ Típicamente “stand-by” o baja corriente

Control PWM modo tensión



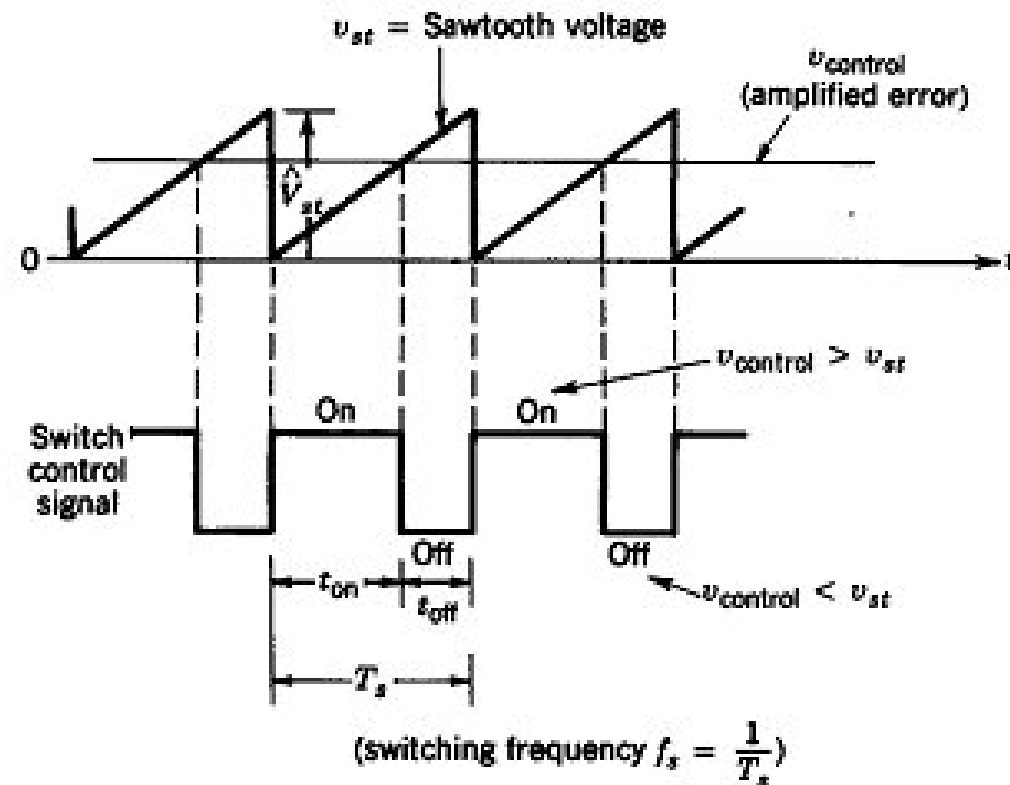
Control PWM modo tensión



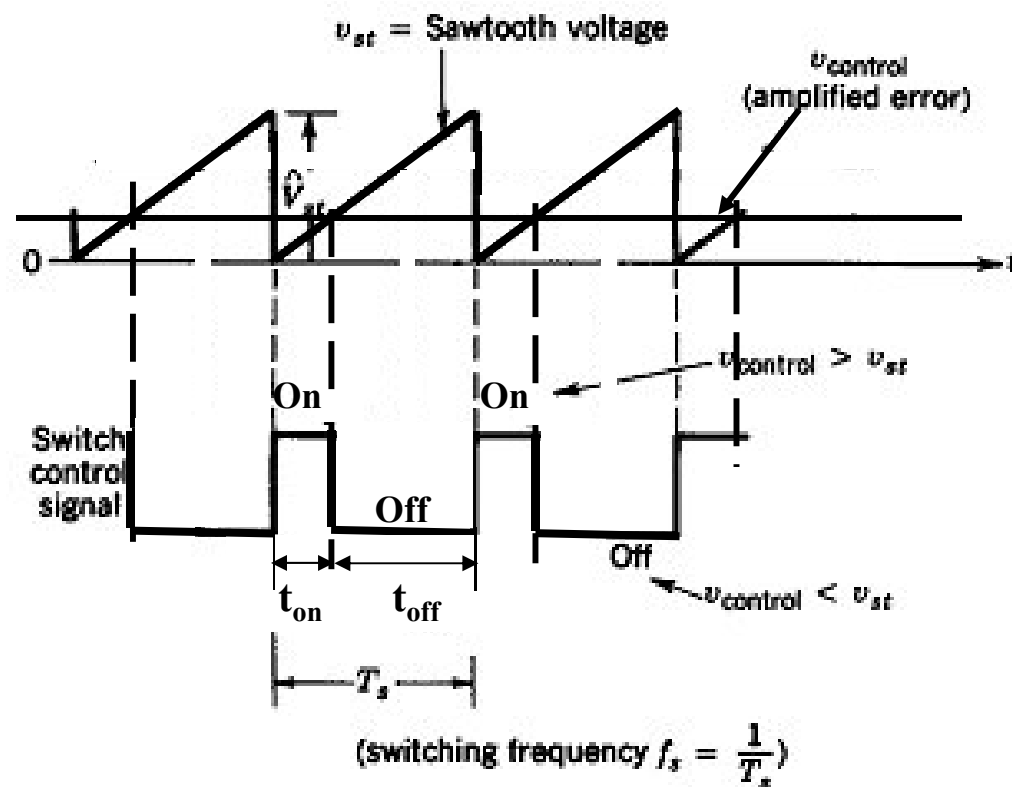
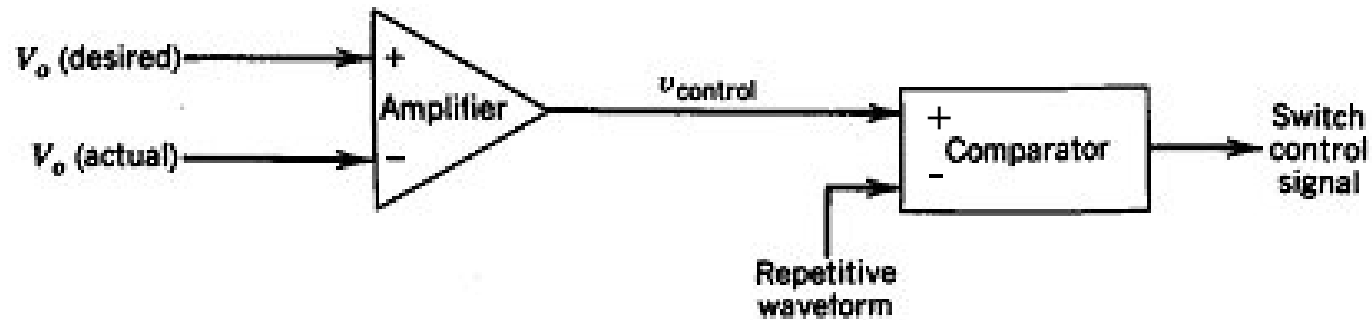
Linealidad
del bloque
comparador

Tª Tales:

$$D = \frac{v_{control}}{\hat{V}_{st}}$$



Control PWM modo tensión



C. I. de control TL-494

SWITCHMODE™ Pulse Width Modulation Control Circuit

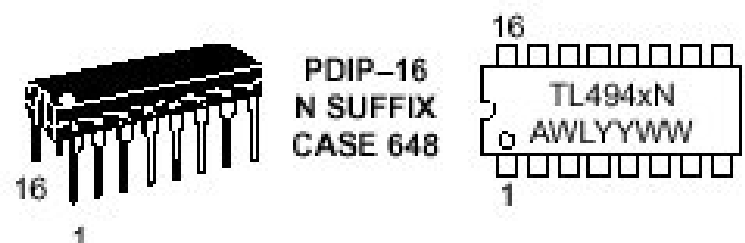
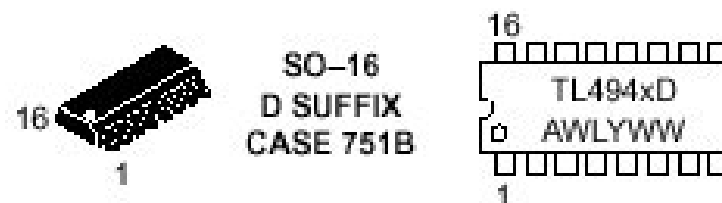
The TL494 is a fixed frequency, pulse width modulation control circuit designed primarily for SWITCHMODE power supply control.

- Complete Pulse Width Modulation Control Circuitry
- On-Chip Oscillator with Master or Slave Operation
- On-Chip Error Amplifiers
- On-Chip 5.0 V Reference
- Adjustable Deadtime Control
- Uncommitted Output Transistors Rated to 500 mA Source or Sink
- Output Control for Push-Pull or Single-Ended Operation
- Undervoltage Lockout

MAXIMUM RATINGS (Full operating ambient temperature range applies, unless otherwise noted.)

Rating	Symbol	Value	Unit
Power Supply Voltage	V_{CC}	42	V
Collector Output Voltage	V_{C1}, V_{C2}	42	V
Collector Output Current (Each transistor) (Note 1)	I_{C1}, I_{C2}	500	mA
Amplifier Input Voltage Range	V_{IR}	-0.3 to +42	V
Power Dissipation @ $T_A \leq 45^\circ\text{C}$	P_D	1000	mW
Thermal Resistance, Junction-to-Ambient	$R_{\theta JA}$	80	$^\circ\text{C/W}$
Operating Junction Temperature	T_J	125	$^\circ\text{C}$
Storage Temperature Range	T_{stg}	-55 to +125	$^\circ\text{C}$
Operating Ambient Temperature Range TL494B	T_A	-40 to +125	$^\circ\text{C}$

MARKING DIAGRAMS



x = B, C or I
A = Assembly Location
WL, L = Wafer Lot
YY, Y = Year
WW, W = Work Week

*This marking diagram also applies to NCV494.

C. I. de control TL-494

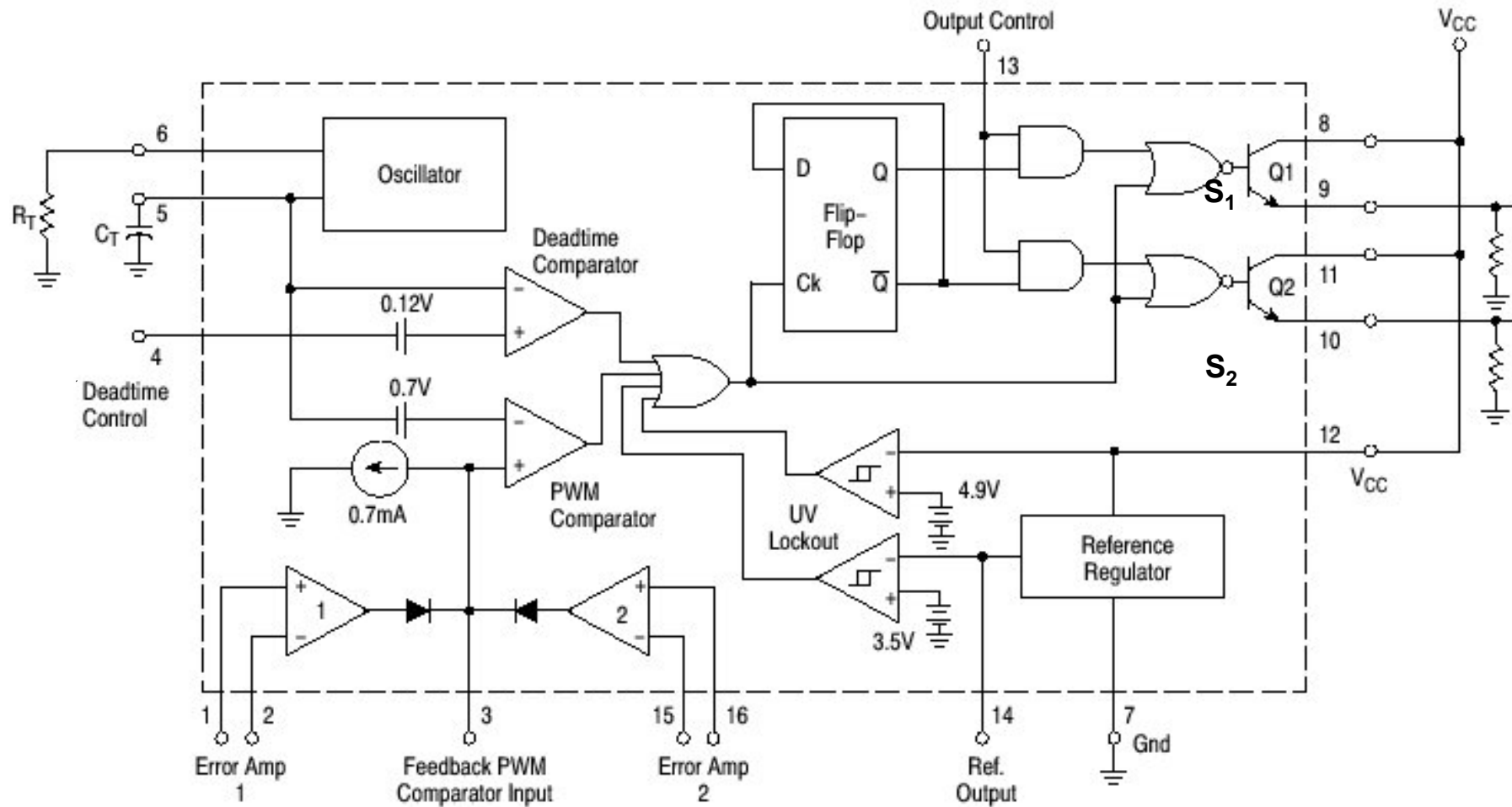


Diagrama de bloques principal

C. I. de control TL-494

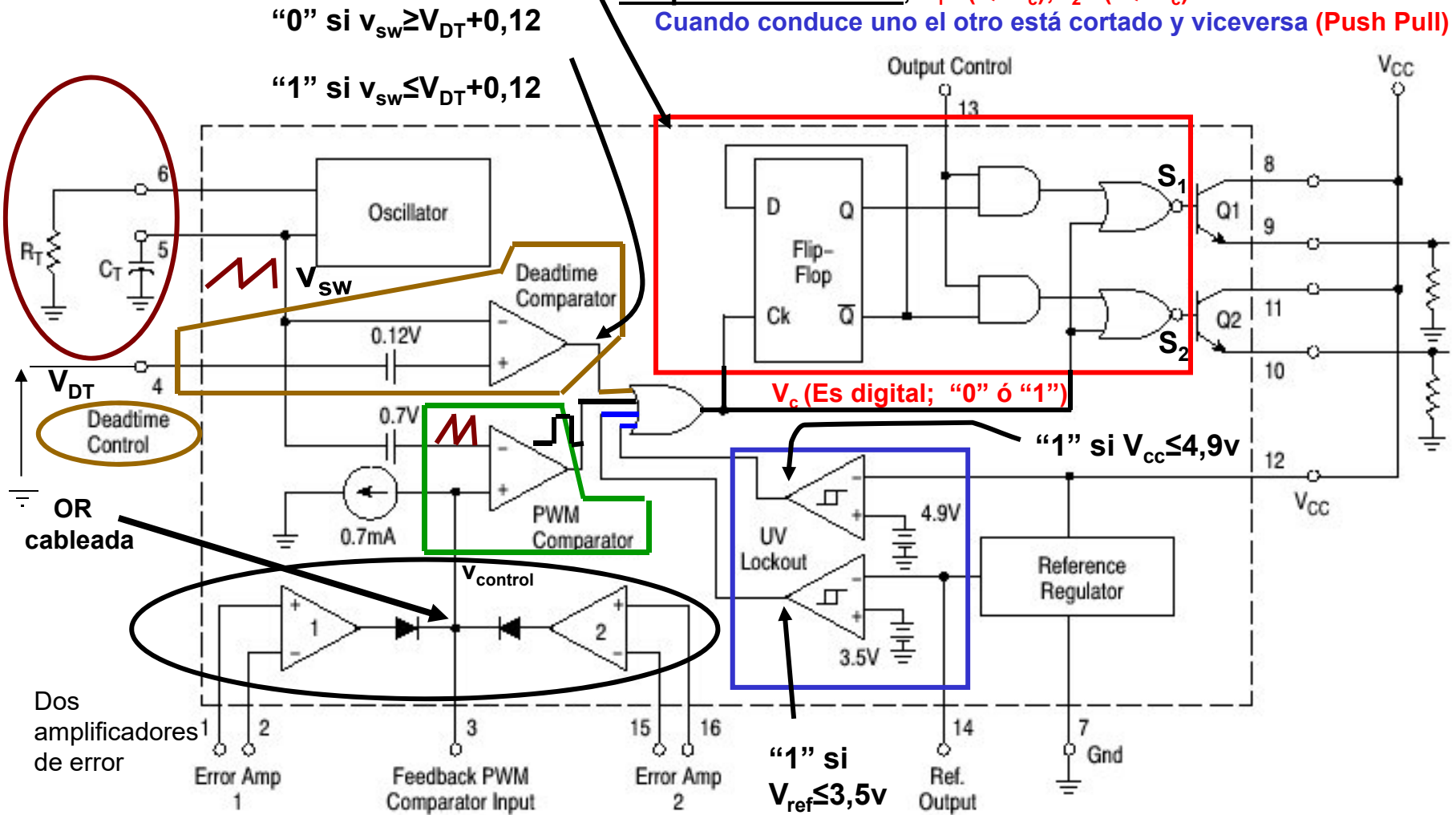
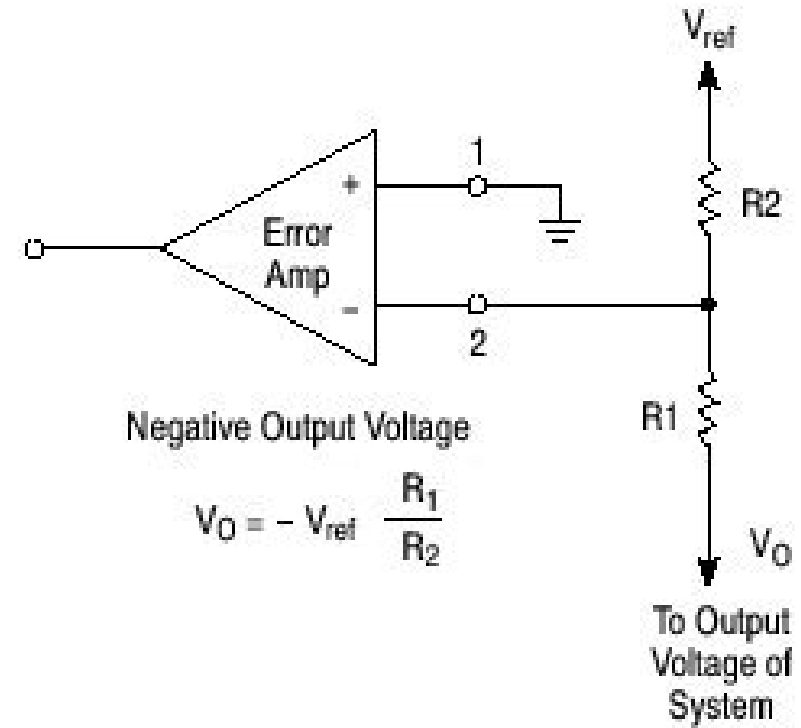
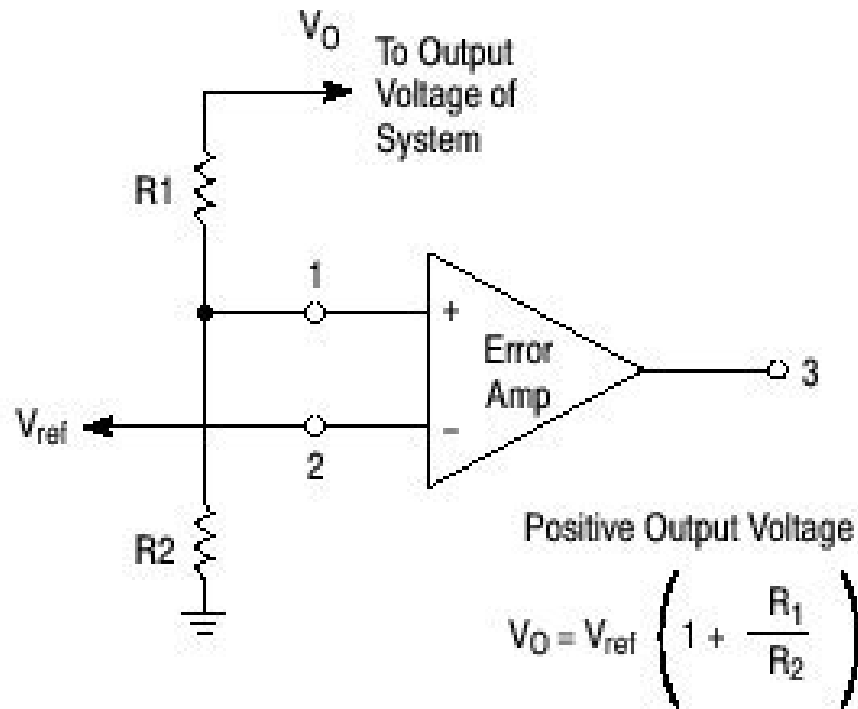


Diagrama de bloques principal

C. I. de control TL-494

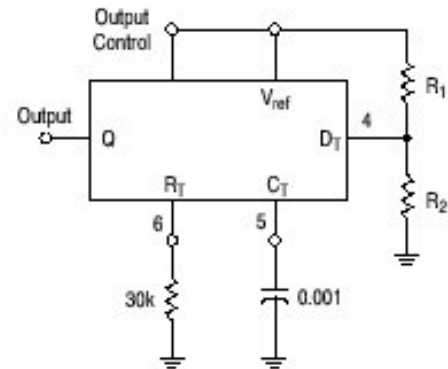


Error-Amplifier Sensing Techniques

ERROR AMPLIFIER SECTION

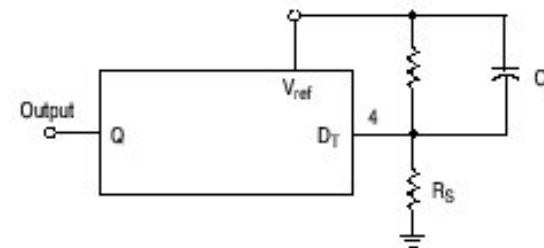
Input Offset Voltage (V_O (Pin 3) = 2.5 V)	V_{IO}	–	2.0	10	mV
Input Offset Current (V_O (Pin 3) = 2.5 V)	I_{IO}	–	5.0	250	nA
Input Bias Current (V_O (Pin 3) = 2.5 V)	I_{IB}	–	–0.1	–1.0	μ A
Input Common Mode Voltage Range ($V_{CC} = 40$ V, $T_A = 25^\circ\text{C}$)	V_{ICR}	–0.3 to $V_{CC}-2.0$			V

C. I. de control TL - 494



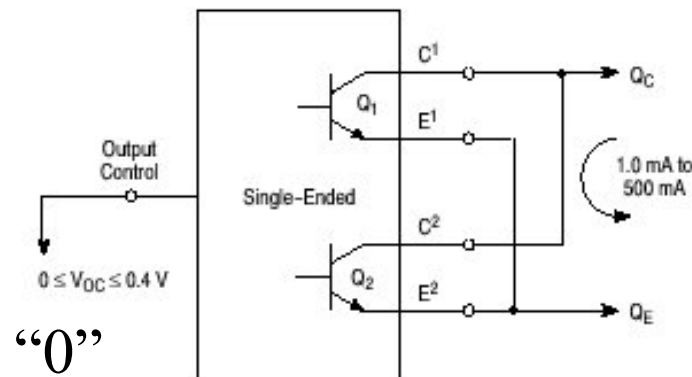
$$\text{Max. \% on Time, each output} = 45 - \left(\frac{80}{1 + \frac{R_1}{R_2}} \right)$$

Deadtime Control Circuit

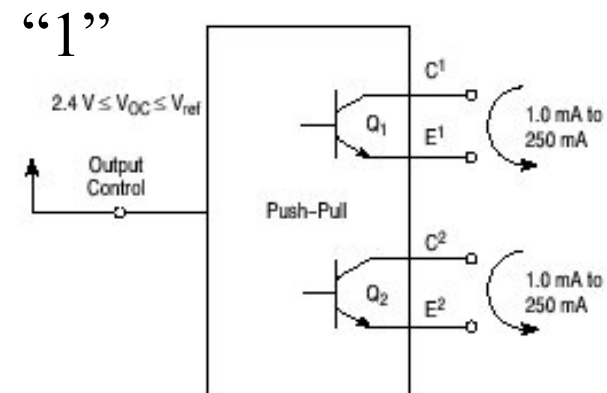


Se evita que Q1 y Q2 conduzcan al encender el chip

Soft-Start Circuit



“0”



“1”

Output Connections for Single-Ended and Push-Pull Configurations

C. I. de control TL-494

Soft Start

Modo Single Ended

(Lógica de control funcionando como inversor)

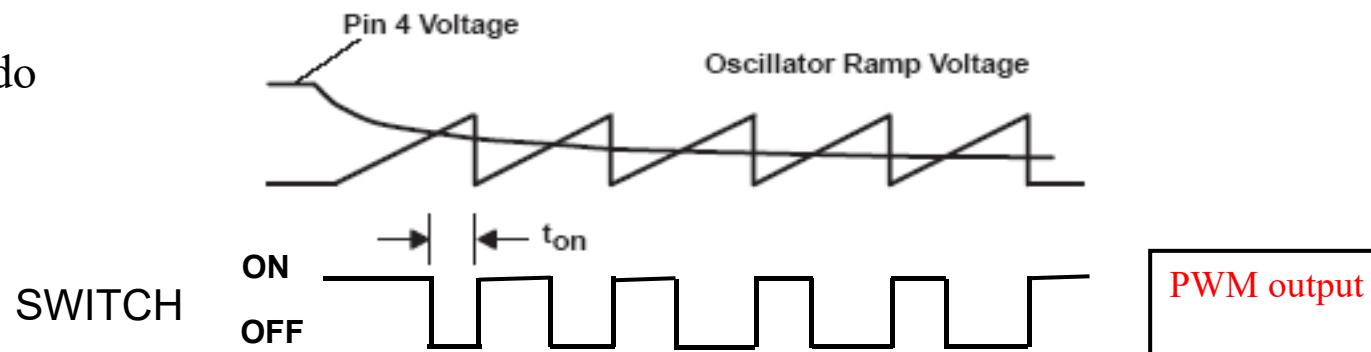
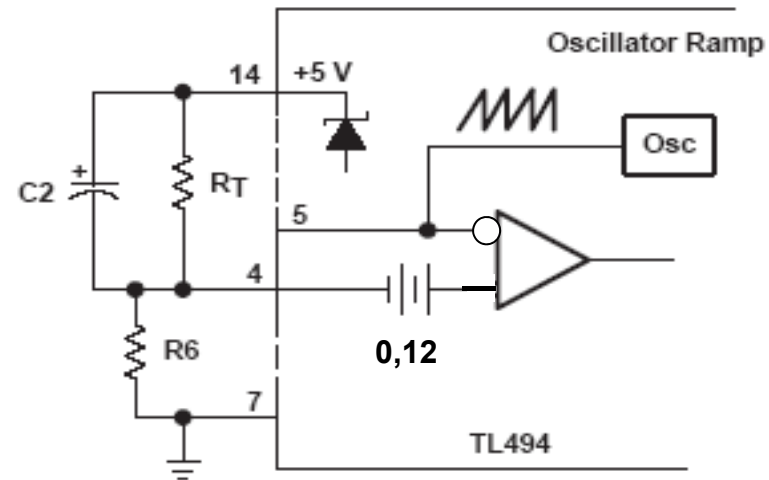
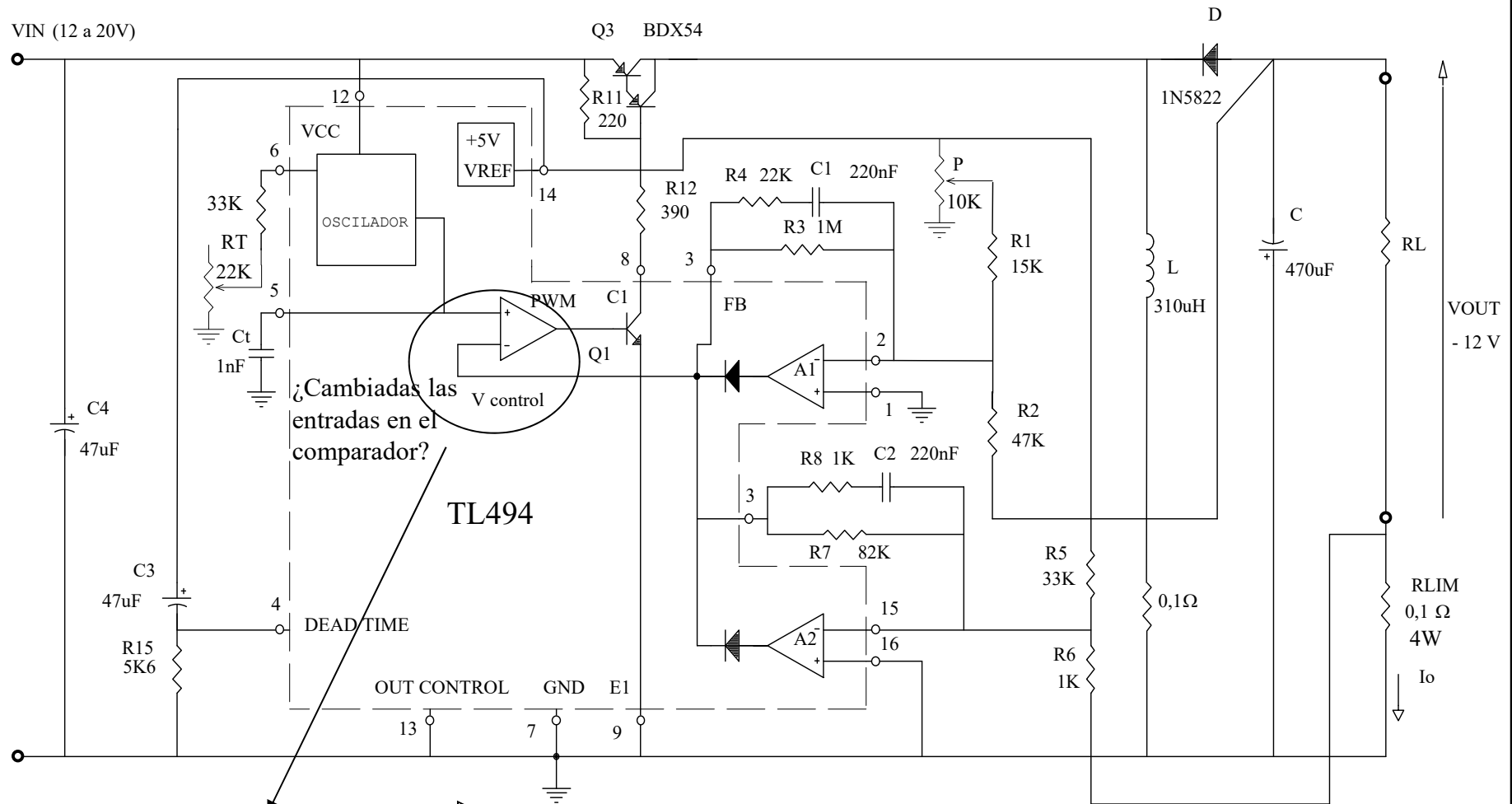


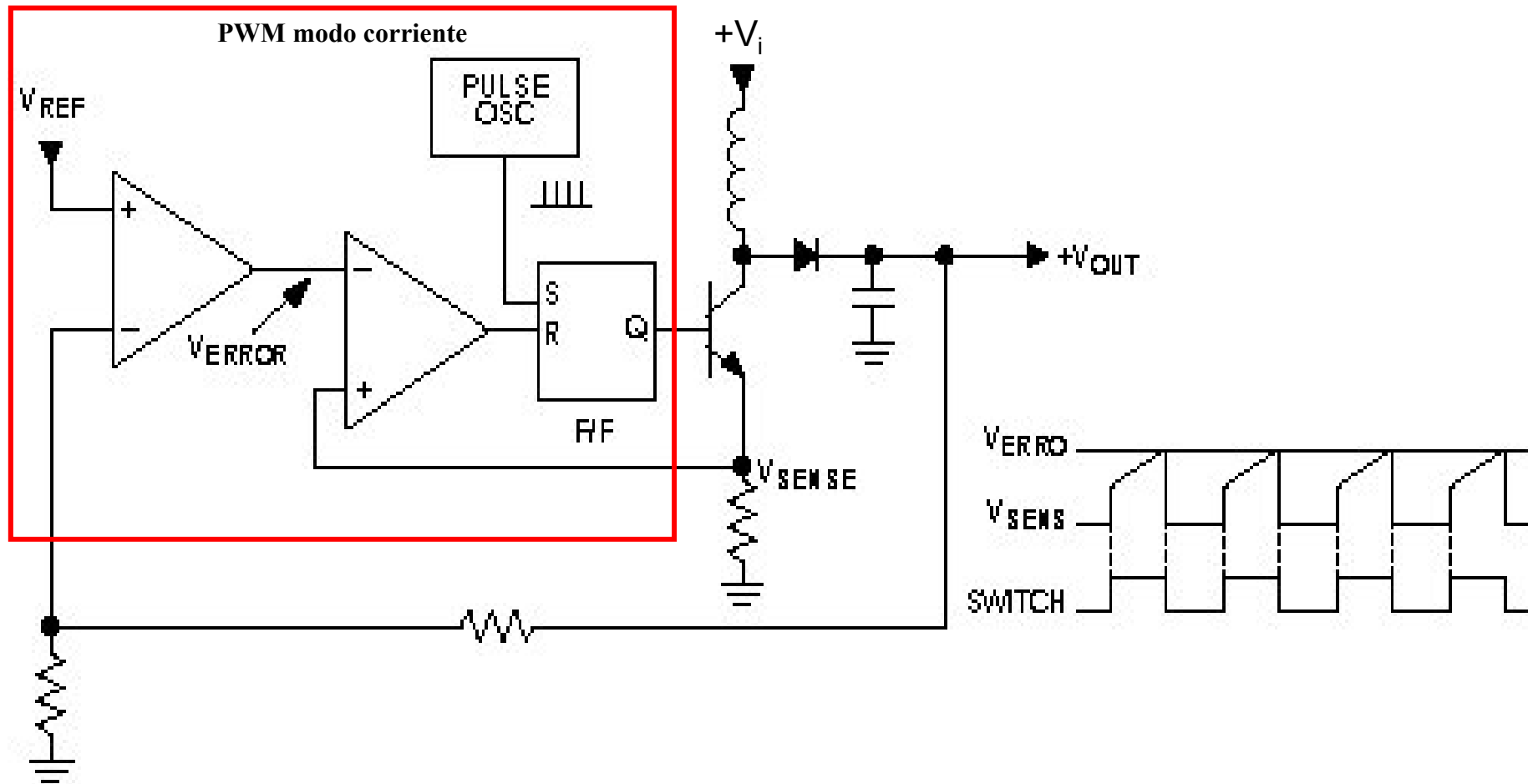
Figure 38. Soft-Start Circuit

Práctica 5

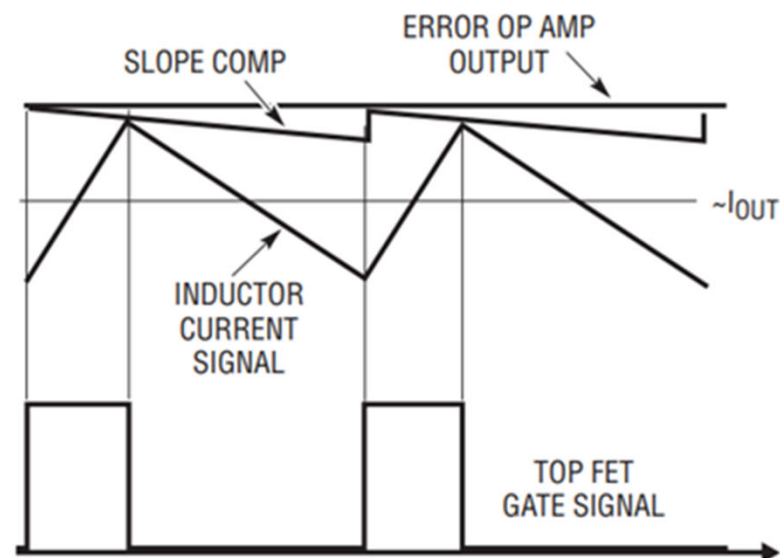
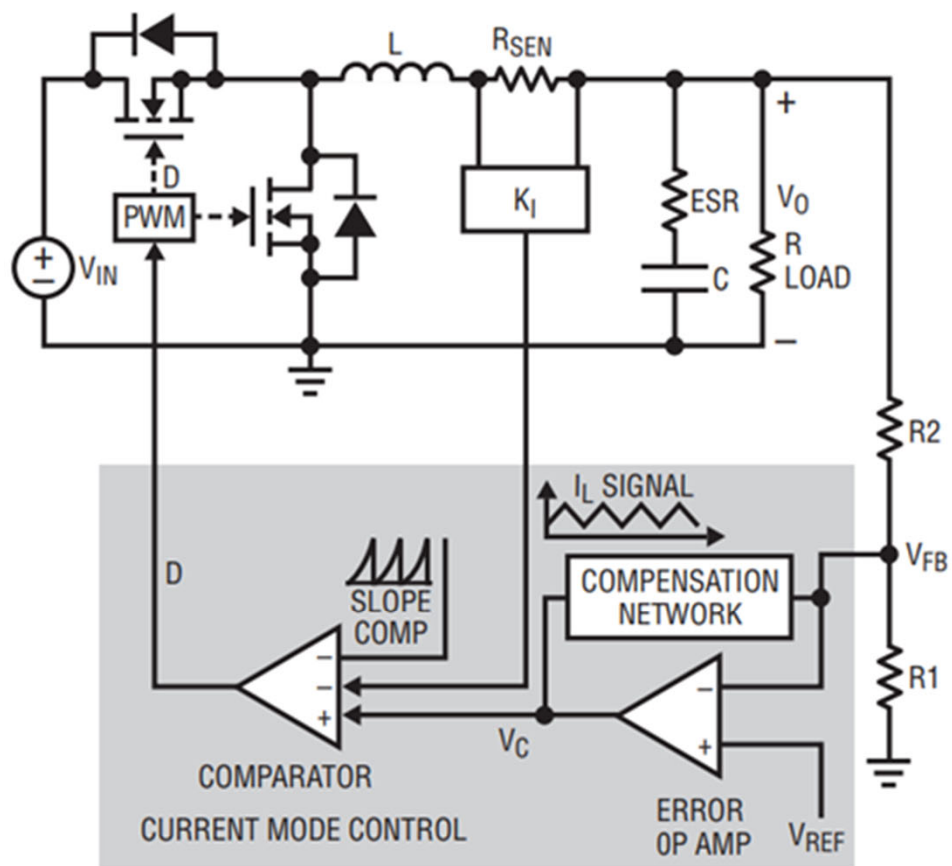


Circuito de aplicación
(simplificado)

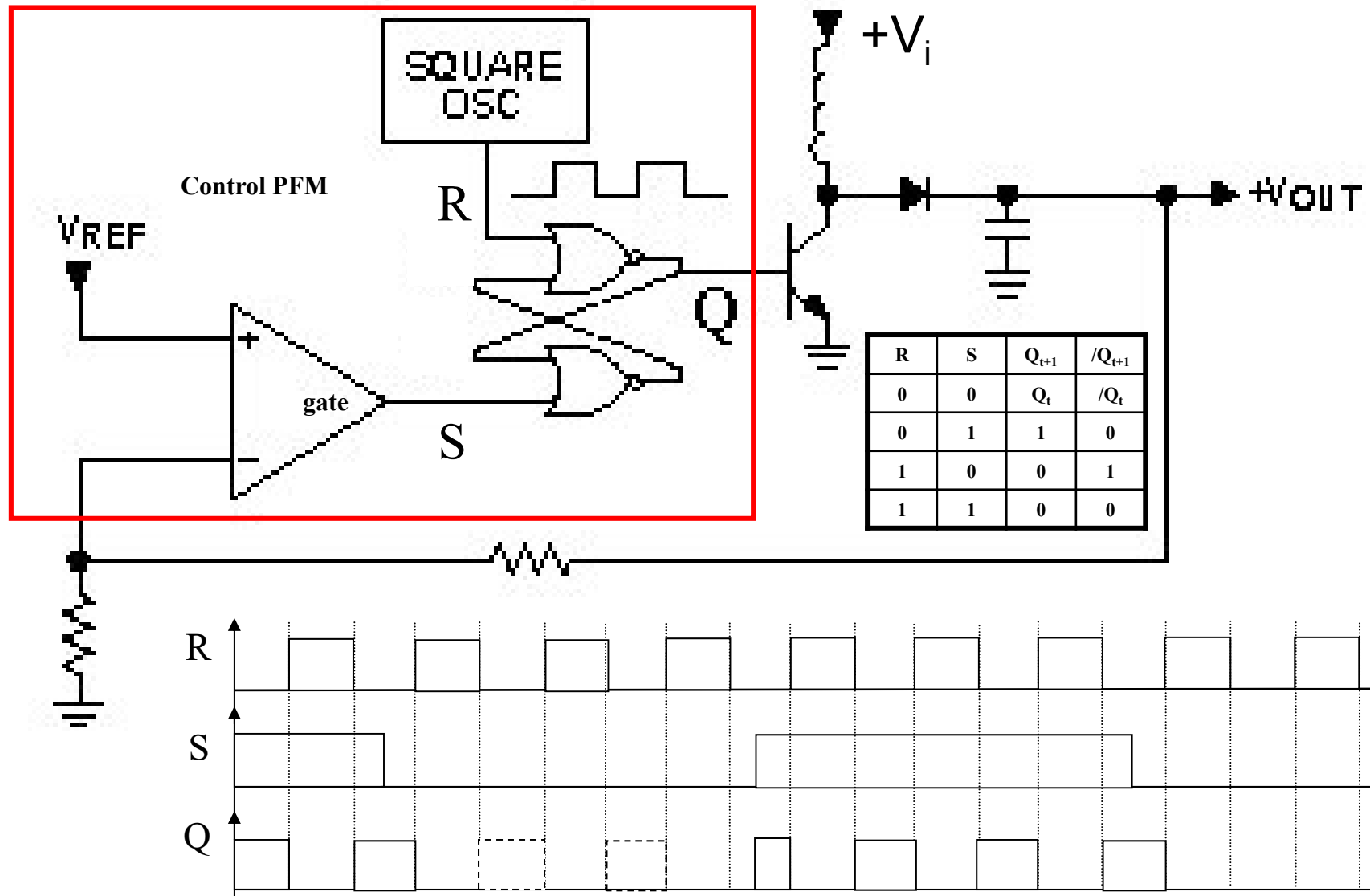
Control PWM modo corriente



Control PWM modo corriente



Control PFM



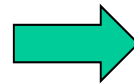
Control: Compensación del amplificador de error

TRABAJANDO EN MODO CONTINUO	Efecto sobre la Función de Transferencia
Todos los convertidores presentan	2 polos (complejos conjugados)
El A. de error introduce	1 polo en B.F.
Las topologías: Elevadora, Elevadora/Reductora, presentan	1 cero en el semiplano derecho (Sistema de fase no mínima, es decir, en la gráfica de la fase este cero se comporta como un polo)
Si se hace control en modo corriente	• Queda 1 polo (real) en la F. de T. de los convertidores. • Además, si $D > 50\%$ se precisa incluir compensación de pendiente (slope compensation) para eliminar oscilación subarmónica

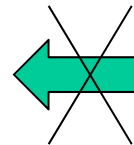
Control: Compensación del amplificador de error

TRABAJANDO EN MODO DISCONTINUO	Efecto sobre la Función de Transferencia
Todos los convertidores presentan	1 polo (real)
El A. de error introduce	1 polo en B.F.
Las topologías: Elevadora, Elevadora/Reductora,	NO presentan el cero en el semiplano derecho

A. de error compensado
en modo continuo



A. de error compensado
en modo discontinuo



No está garantizado

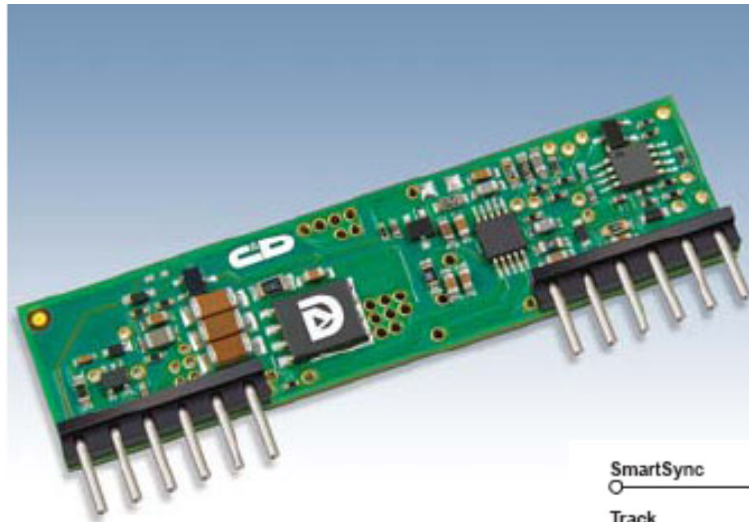
Modo y control recomendado para cada convertidor

Tipo de convertidor	Modo	Control
Reductor	Continuo	Corriente
Elevador	Discontinuo	Corriente o Feed forward
Elevador/ Reductor	Discontinuo	Corriente o Feed forward

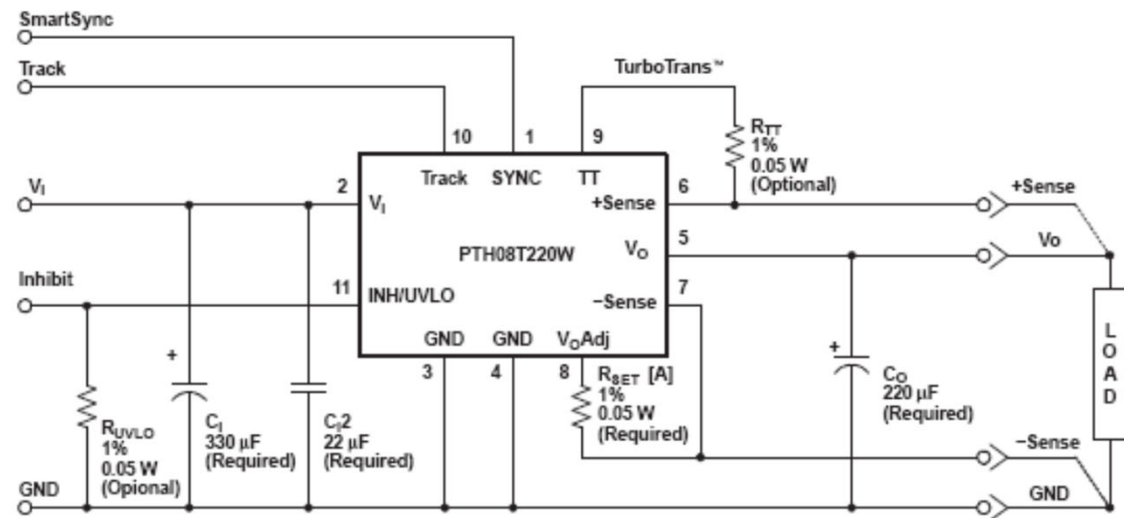
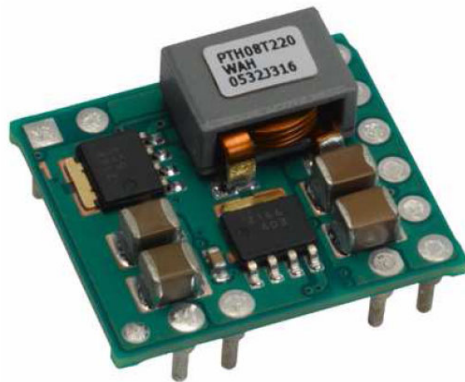
Convertidores DC-DC integrados, PoL converters

- Los sistemas electrónicos actuales están alimentados con bajas tensiones y demandan corrientes elevadas (Ej. Microprocesador de un ordenador)
- Los “PoL converters” (Point-of-Load power supplies) cumplen estas especificaciones.
- Los “PoL converters” se encuentran normalmente en el mercado como módulos de inserción realizados con tecnología SMT, que pueden integrarse en una PCB como un componente más.
- Los fabricantes se han puesto de acuerdo para que los módulos de unas determinadas características, tengan las mismas dimensiones y configuración de pines, independientemente del fabricante con los componentes PoLA = PoL Alliance.

Convertidores DC-DC integrados, PoL converters

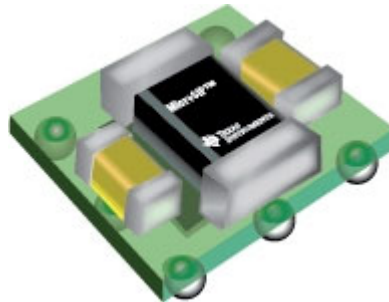


- Sin aislamiento
- Encapsulado según estándar industrial DOSA y configuración de pines
- Amplio rango de tensión de entrada
- Corriente de salida máxima 6, 10 o 16 A
- Tensión de salida seleccionada por usuario mediante resistencias externas
- Inicio por fases para secuenciación y seguimiento
- Control de encendido/apagado remoto
- Detección remota
- Bajo rizado y ruido de salida
- Bloqueo subtensión de entrada, protección temperatura excesiva y cortocircuito continuo



Convertidores DC-DC integrados

600-mA, HIGH-EFFICIENCY MicroSiP™ STEP-DOWN CONVERTER (PROFILE <1.0mm)



FEATURES

- 90% Efficiency at 5.5MHz Operation
- 17 μ A Quiescent Current
- Wide V_{IN} Range From 2.3V to 4.8V
- 5.5MHz Regulated Frequency Operation
- Spread Spectrum, PWM Frequency Dithering
- *Best in Class* Load and Line Transient
- $\pm 2\%$ Total DC Voltage Accuracy
- Automatic PFM/PWM Mode Switching
- Low Ripple Light-Load PFM Mode
- ≥ 35 dB V_{IN} PSRR (1kHz to 10kHz)
- Internal Soft Start, 120- μ s Start-Up Time
- Integrated Active Power-Down Sequencing (Optional)
- Current Overload and Thermal Shutdown Protection
- Sub 1-mm Profile Solution
- Total Solution Size <6.7 mm²

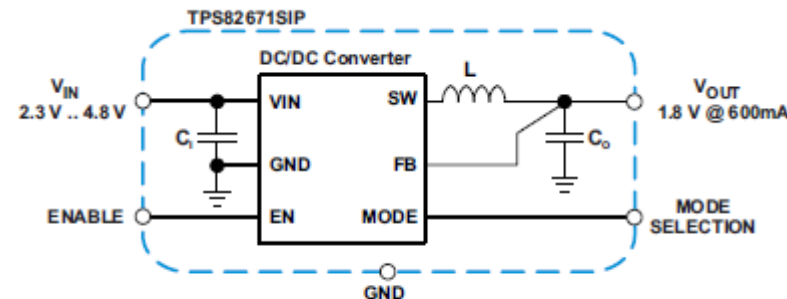


Figure 1. Typical Application

APPLICATIONS

- Cell Phones, Smart-Phones
- Digital TV, WLAN, GPS and Bluetooth™ Applications
- POL Applications

SLVSAI0F –OCTOBER 2010–REVISED NOVEMBER 2012

Convertidores DC-DC integrados

4.5-V to 14.5-V Input, 6-A Synchronous Buck, Integrated Power Solution



9 mm × 15 mm × 2.8 mm

APPLICATIONS

- Broadband & Communications Infrastructure
- Automated Test and Medical Equipment
- Compact PCI / PCI Express / PXI Express
- DSP and FPGA Point of Load Applications
- High Density Distributed Power Systems

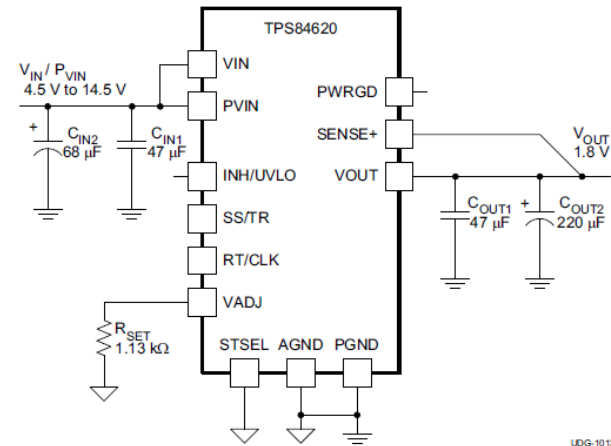
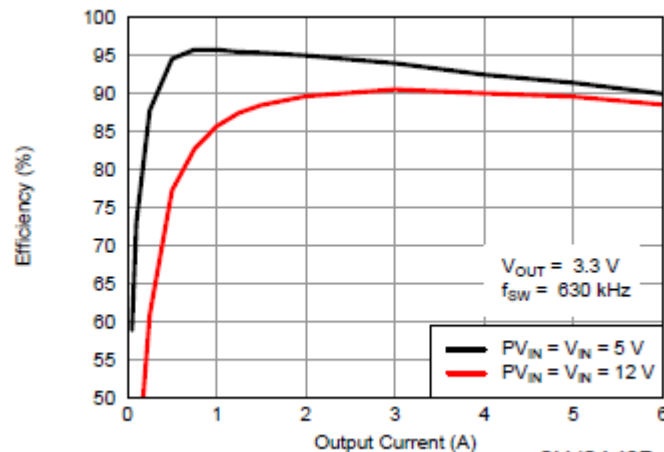


Figure 25. Typical Schematic
 $PV_{IN} = V_{IN} = 4.5 \text{ V to } 14.5 \text{ V}$, $V_{OUT} = 1.8 \text{ V}$

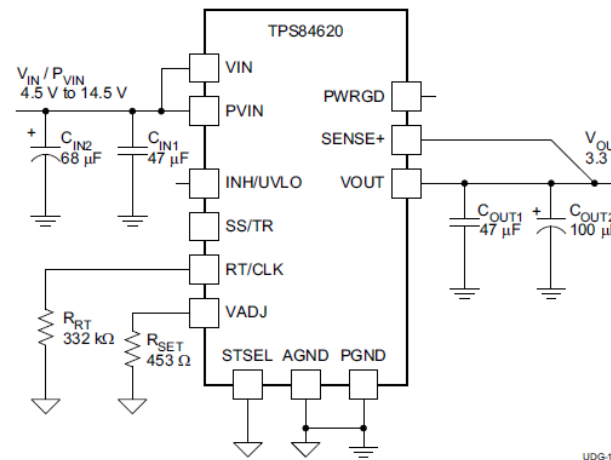


Figure 26. Typical Schematic
 $PV_{IN} = V_{IN} = 4.5 \text{ V to } 14.5 \text{ V}$, $V_{OUT} = 3.3 \text{ V}$

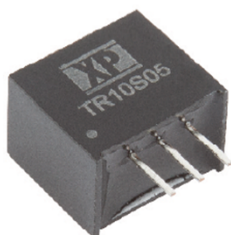
Convertidores DC-DC integrados

Standard Application

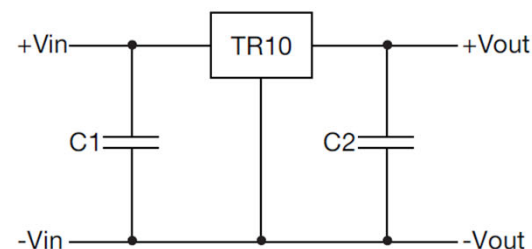
TR10 Series

1 Amp

- 3 Pin Switching Regulator
- SIP Package
- -40 °C to +85 °C Operation
- Full Load to 60 °C Ambient
- Wide Input Range to 28 V
- Class B Conducted & Radiated Emissions
- MTBF >3.8 Mhrs
- 3 Year Warranty

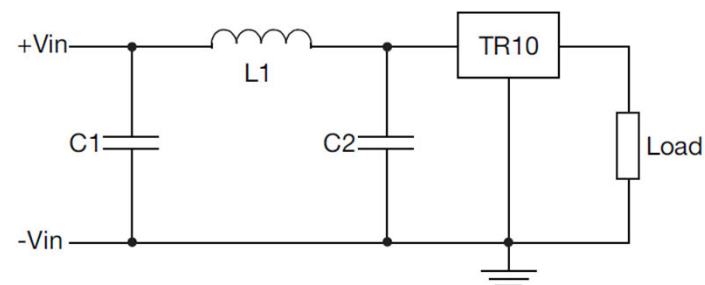


EMI Filter



C1 = 22 μ F

C2 = 47 μ F (optional) to improve transient response



C1 = 10 μ F, 50 V

L1 = 22 μ H

C2 = 10 μ F, 50 V

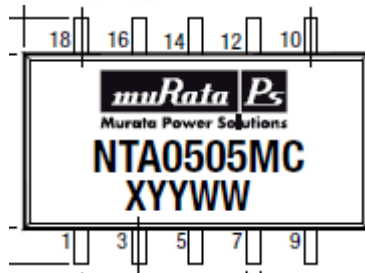
C1, C2 and L1 should be placed as close to the TR10 as possible

Models & Ratings

Input Voltage	Output Voltage	Output Current	Input Current			Max. Capacitive Load	Efficiency	
			No Load	Full Load, min. Vin	Full Load, max. Vin		Min. Vin	Max. Vin
7-28 V	3.3 V	1000 mA	1.5 mA	530 mA	145 mA	220 μ F	89%	82%
8-28 V	5.0 V	1000 mA	1.5 mA	670 mA	210 mA	220 μ F	93%	86%

Convertidores DC-DC integrados aislados

Isolated 1W Dual Output SM DC/DC Converters



Pin	Function
1	-V _{IN}
3	+V _{IN}
5	NA
7	OV
9	-V _{OUT}
10	NA
12	+V _{OUT}
14	NA
16	NA
18	NA

SELECTION GUIDE

Order Code ¹	Nominal Input Voltage	Output Voltage	Output Current	Input Current at Rated Load	Efficiency	Isolation Capacitance	MTTF ²
	V	V	mA	mA	%	pF	kHrs
Recommended In Production							
NTA0505MC	5	±5	±100	290	69	33	1697
NTA0509MC	5	±9	±55	267	75	38	682
NTA0512MC	5	±12	±42	260	77	44	343
NTA0515MC	5	±15	±33	256	78	43	188
NTA1205MC	12	±5	±100	121	69	50	559
NTA1212MC	12	±12	±42	111	75	89	243
NTA1215MC	12	±15	±33	110	76	100	154

Maxi, Mini, Micro DC-DC Converters

BRICK

Overview

- Broadest selection of DC-DC modules in the industry
- Input voltages: 9 – 425 Vdc
- Output voltages: 2 – 54 Vdc
- Output power: Maxi (160 – 600 W)
Mini (100 – 300 W)
Micro (50 – 150 W)
- 100°C baseplate operation, no de-rating
- 3,000 Vac isolation
- Agency approvals: cURus, cTUVus, CE Mark



Maxi – full brick
4.60 x 2.2 x 0.5in
[117 x 55,9 x 12,7mm]



Mini – half brick
2.28 x 2.2 x 0.5in
[57,9 x 55,9 x 12,7mm]



Micro – quarter brick
2.28 x 1.45 x 0.5in
[57,9 x 36,8 x 12,7mm]



Features

- Robust packaging for harsh environments
- Extended temperature range (-55 to 100°C)
- Single-wire paralleling
- Output voltage adjust 10 – 110%
- Logic enable/disable
- Input undervoltage lockout
- Output overvoltage protection
- Overtemperature shutdown

Rev 1.5

01/2018